

CONTRIBUTING FACTORS TO FLUX CHANGES IN NONRADIAL STELLAR PULSATIONS

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Abstract. The linearized analytic representation of observed flux changes due to nonradial pulsations is examined under conditions applicable to a variety of pulsating star types. Histograms are presented as a function of l , documenting the relative importance of local temperature, geometry, pressure, and limb-darkening variations, which are all contributing factors to the predicted flux changes. The most favourable situation for light and colour data to assist in mode discrimination for low l requires, amongst other things, that the geometry-related factor be significant. This is noted to occur only if the Q value of the pulsation is not grossly different from the Q value of a radial fundamental pulsation. Published light and colour data for β Cephei, 53 Persei, δ Scuti, rapidly oscillating Ap, Cepheid, and ZZ Ceti variables are compared with model predictions on an amplitude ratio versus phase difference plane ($A_{\text{col}}/A_{\lambda_1}$, $\phi_{\text{col}} - \phi_{\lambda_1}$). With the notable exception of the rapidly oscillating Ap stars, these comparisons suggest consistency of the nonradial flux change model with other known constraints on the different variable star groups.

1. Introduction

Nonradial stellar pulsation is known to occur in a wide variety of stellar types. Identification of nonradial stellar pulsation in hot stars above the Main Sequence extends to the β Cephei variables and the surrounding group of 53 Persei variables (Smith, 1981; Smith *et al.*, 1984), the extreme helium stars such as BD – 9°4395 (Jeffery *et al.*, 1985) and the Be stars (Baade, 1986). In cooler systems above the Main Sequence, such identification has been made in some δ Scuti stars (Breger, 1980; Smith, 1982) and for the rapidly oscillating Ap stars (Kurtz, 1986). Small scale nonradial pulsation in the Sun has been clearly demonstrated (Toomre, 1986) and examination of other stars for a similar phenomenon is being undertaken (Gough, 1985). Below the Main Sequence, the variable DA white dwarfs, or ZZ Ceti stars, and the variable DB white dwarfs, or GD358 stars, are known nonradial pulsators (Winget and Fontaine, 1982; Liebert *et al.*, 1986). More recently, a group of very hot degenerate variables, the PG1159–035 stars, has also been shown to be nonradial pulsators (Bond *et al.*, 1984). Representative locations of some variable star regimes on the (T_e , $\log g$) plane are given in Figure 1.

Three techniques for the identification of nonradial stellar pulsation are employed. The first and most important of these is the determination of the frequency spectrum of observed oscillations. This also provides a tool for investigating internal structure. The second technique relies on observations of spectral line profile variations. The third approach requires an examination of the amplitude and phase of continuum flux variations. Aspects of this latter approach are the subject of this paper.

The first formulation of an analytic expression for the flux variations due to small amplitude nonradial pulsations was presented by Dziembowski (1977). Subsequently

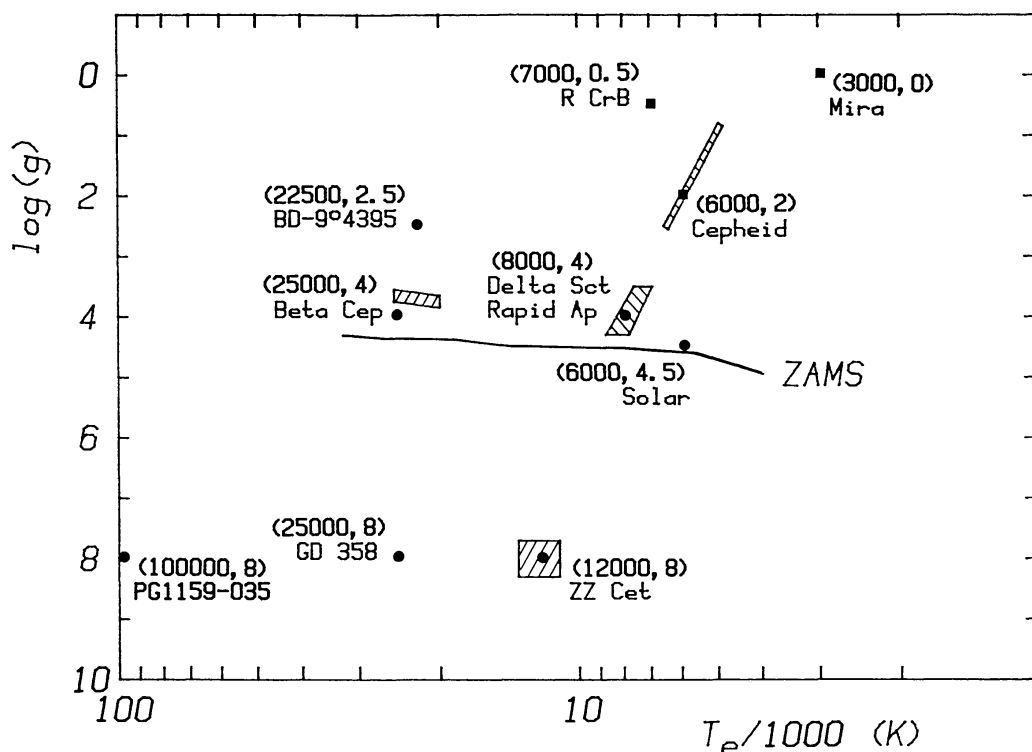


Fig. 1. Selected variable star regimes on the $(T_e, \log g)$ plane. Hatched areas indicate the observed instability regions of the β Cephei, δ Scuti, Cepheid, and ZZ Ceti variables. The specified $(T_e, \log g)$ of regimes are representative points only, selected for convenient use of published stellar model atmosphere data. Filled squares denote regimes where only radial pulsations are known. Filled circles denote regimes where at least some variables are suspected to be nonradial pulsators.

this formulation was expanded upon by Buta and Smith (1979) in relation to the 53 Persei stars, by Balona and Stobie (1979b, 1980a) and Stamford and Watson (1981) in connection with the β Cephei and δ Scuti variables, and by Robinson *et al.* (1982) in a discussion of the ZZ Ceti stars. Also Balona (1981) has detailed the effect of gravity changes on the observed flux variations.

Here a formulation of the analytic expression for flux variations is given (Section 2), together with a discussion of the sources of necessary stellar atmosphere parameters and other input data required for the model (Section 3). The model is then used in a systematic way in Section 4 to document the relative importance of the different terms contributing to predicted flux changes in several test regimes applicable to various pulsating star types. A general rule is presented to indicate under what circumstances light and colour data are most likely to be useful for low l -mode discrimination. Finally, in Section 5, relevant published light and colour data for pulsating variable types are surveyed and compared directly with the model to demonstrate its applicability and current limitations.

2. Linearized Flux Variations

In the linearized regime, the predicted flux change for a nonradial pulsation with mode indices (l, m) is, in magnitudes,

$$\Delta m(\lambda, t) = -1.086\varepsilon P_{l, |m|}(\mu_0) ((T_1 + T_2) \cos(\omega t + \psi_T) + (T_3 + T_4 + T_5) \cos(\omega t)), \quad (1)$$

where ε is an amplitude parameter and μ_0 is the cosine of the inclination of the stellar pulsation axis to the observer. The terms T_1 and T_3 account, respectively, for the local temperature changes on the stellar surface and the local geometry changes. By use of notation employed by Stamford and Watson (1981), they may be expressed as

$$T_1 = b_{l\lambda} \alpha_T(\lambda) B \quad (2)$$

and

$$T_3 = b_{l\lambda} (2 + l) (1 - l). \quad (3)$$

The T_4 term attempts to account for local surface pressure changes, which in some circumstances may be significant. By following Stamford and Watson, it may be written as

$$T_4 = -b_{l\lambda} p^* C \alpha_g(\lambda). \quad (4)$$

The terms T_2 and T_5 permit allowance to be taken of any variation in limb-darkening properties as local $(T_e, \log g)$ values change on the stellar surface during the pulsation. They are

$$T_2 = \frac{\partial b_{l\lambda}}{\partial \log T_e} \frac{B}{2.3026} \quad (5)$$

and

$$T_5 = -\frac{\partial b_{l\lambda}}{\partial \log g} \frac{p^* C}{2.3026}. \quad (6)$$

These T_2 and T_5 terms have not been included in previous discussions of flux variations in nonradial pulsations. They are generally not important if discussion is restricted to modes with $l \leq 2$. However, as will be seen in Section 4, they may be quite significant if l values greater than this are considered. Their derivation follows from linearizing the normalized limb-darkening function $h_\lambda(\mu)$ with respect to $(T_e, \log g)$ changes, and incorporating this into the analytic reduction originally outlined by Dziembowski (1977).

The $b_{l\lambda}$ in (2) to (4) is the weighted limb-darkening integral

$$b_{l\lambda} = \int_0^1 h_\lambda(\mu) \mu P_l(\mu) d\mu. \quad (7)$$

Temperature and gravity derivatives of this appear in (5) and (6). The $\alpha_T(\lambda)$ and $\alpha_g(\lambda)$ in (2) and (4) are local flux derivatives obtainable from stellar model atmospheres. They have the form

$$\alpha_T(\lambda) = \frac{\partial \log F_\lambda}{\partial \log T_e} \quad (8)$$

and

$$\alpha_g(\lambda) = \frac{\partial \log F_\lambda}{\partial \log g} . \quad (9)$$

The p^* occurring in (4) and (6) is a measure of the variation of atmospheric gas-pressure with surface gravity in layers giving rise to the observed continuum flux. Its form is

$$p^* \approx \left(\frac{\partial \log g}{\partial \log p_g} \right)_{\tau=1} . \quad (10)$$

Also in (4) and (6) we have

$$C = (4 + \alpha_H^{-1}) - l(l+1)\alpha_H, \quad (11)$$

where

$$\alpha_H = G\rho_\odot Q^2/3\pi;$$

and Q is the stellar pulsation constant. This is the boundary condition discussed by Buta and Smith (1979). Here the C value is the negative of the ratio of local fractional gas-pressure amplitude to local fractional radius amplitude. C is always positive for radial pulsation but may be either positive or negative for nonradial pulsation. In the nonradial case, a positive C indicates a dominance of the radial component of the surface motion. Thus, regions of local contraction have the highest pressures and, in the adiabatic limit, also the highest effective temperatures. A negative C indicates the dominance of horizontal motions on the stellar surface. In this case, the locally contracted regions have lowest pressures, because material has flowed horizontally away from these regions.

Finally, the parameter B appearing in (2) and (5), which is a measure of the ratio of local fractional temperature amplitude to local fractional radius amplitude, will be here written in terms of a positive quantity R , defined by

$$B = RB_{\text{adiab}} = R(1 - \Gamma_2^{-1})C. \quad (12)$$

This R , together with ψ_T in (1), is used to describe departures from adiabatic conditions on the stellar surface. In the adiabatic situation $(R, \psi_T) = (1, \pi)$. In the non-adiabatic situation, R and $(\psi_T - \pi)$ specifically give the departures in amplitude and phase of the local fractional temperature change from its adiabatic amplitude and phase, as measured with respect to local fractional gas-pressure change. The latter is given by

$$(\Delta T_e/T_e)_{\text{adiab}} = (1 - \Gamma_2^{-1})\Delta p_g/p_g. \quad (13)$$

3. Parameters in the Model

In the evaluation of $\Delta m(\lambda, t)$ using (1), the quantities $\alpha_T(\lambda)$, $\alpha_g(\lambda)$, p^* , and $b_{l\lambda}$, together with the temperature and gravity derivatives of $b_{l\lambda}$, may all be obtained from stellar model atmospheres at the $(T_e, \log g)$ of the star or group of stars in question, provided suitable published data exist. For a given mode (l, m) , the remaining unknown quantities are ε , μ_0 , Q , Γ_2 , R , and ψ_T . In this paper the unknowns ε and μ_0 , together with m , are not required. In first instance this is because only the relative values of the components T_1 and T_5 are discussed in Section 4. In second instance it is because colour data are used in Section 5, when comparing observations with the model in an attempt to determine l , thus obviating the necessity to know $[\varepsilon P_{l, |m|}(\mu_0)]$.

The following subsections discuss generation of the necessary stellar atmosphere parameters and estimation of the quantities Q , Γ_2 , and (R, ψ_T) for the variable star test regimes subsequently examined.

3.1. FLUX DERIVATIVES

Fluxes of Kurucz (1979) enable $\alpha_T(\lambda)$ and $\alpha_g(\lambda)$ to be generated over a wide range of $(T_e, \log g)$ for UBV and $uvby$ filters. At cooler temperatures these may be supplemented by reference to Bell and Gustafsson (1978). Bell and Gustafsson do not list directly any fluxes as a function of $(T_e, \log g)$, which is what is required. They do, however, specify the $(T_e, \log g)$ dependence of the bolometric correction, and this suffices to determine $\alpha_T(V)$ and $\alpha_g(V)$ indirectly, since

$$\alpha_T(V) = 4 + 0.4 \frac{\partial \text{Bol. Cor.}}{\partial \log T_e} \quad (14)$$

and

$$\alpha_g(V) = 0.4 \frac{\partial \text{Bol. Cor.}}{\partial \log g} \quad (15)$$

Once $\alpha_T(V)$ and $\alpha_g(V)$ have been obtained, values for other filter bands follow from tables of colour dependence on $(T_e, \log g)$.

Figures 2 and 3 show, as examples, plots of $\alpha_T(V)$, $\alpha_T(B)$, and $\alpha_g(V)$ for solar abundance atmospheres. The sensitivity of the flux derivatives to errors in the adopted $(T_e, \log g)$ of a star depends on the filter band. For the V band, the sensitivity rises markedly for T_e cooler than about 9000 K. Table I gives values of $\alpha_T(\lambda)$ and $\alpha_g(\lambda)$ for $(T_e, \log g)$ regimes of some variable star types. Accuracy of reading from the models is typically ± 0.1 for $\alpha_T(\lambda)$ and ± 0.002 to ± 0.004 for $\alpha_g(\lambda)$, but considerably poorer in the Mira case where temperature extrapolation is required. For this case, accuracy is probably only ± 1 in $\alpha_T(\lambda)$ and ± 0.02 in $\alpha_g(\lambda)$. Adopted $(T_e, \log g)$ are intended as representative values only, selected for convenient use of model atmosphere data. The 53 Persei case was also selected with a specific data application in mind (Section 5.2). No adequate data exist for BD-9°4395, rapid Ap, R CrB, and PG1159-35 regimes because of abundance peculiarities.

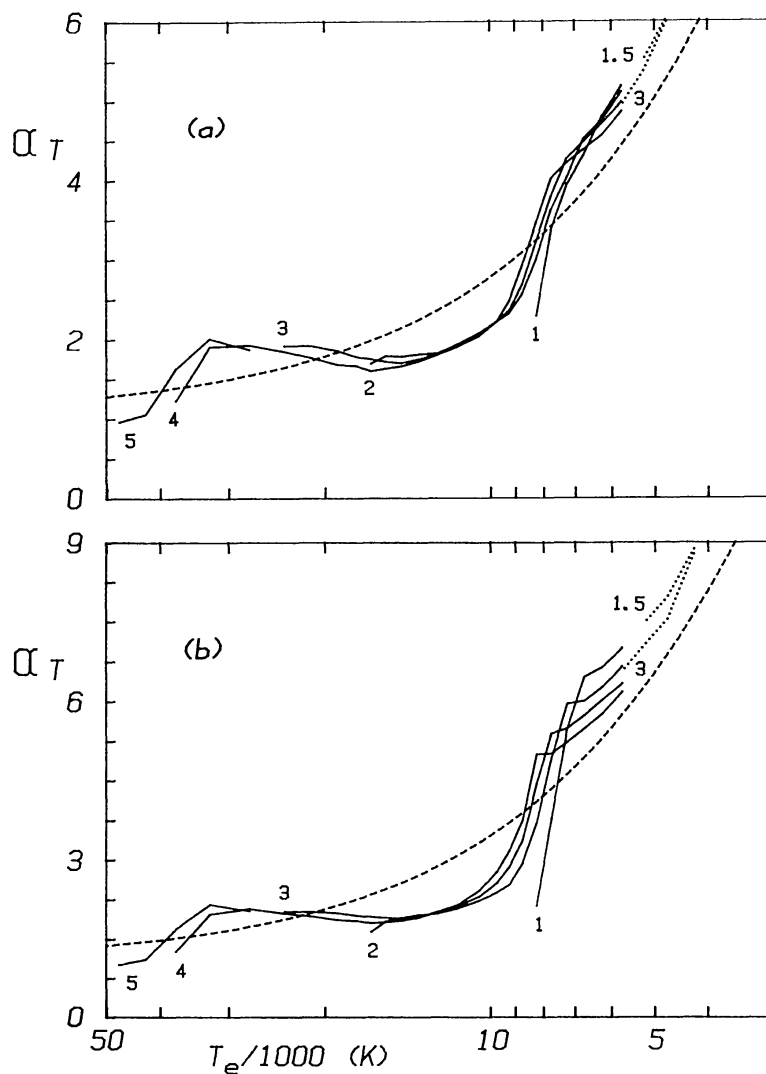


Fig. 2. Dependence of the flux derivative $\alpha_T(\lambda)$ on $(T_e, \log g)$ for solar abundance atmospheres. Data derived from Kurucz (1979) are shown as full lines and those derived from Bell and Gustafsson (1978) are shown as dotted segments. The $\log g$ values are marked. Panel (a) is for V band data and panel (b) is for B band data. Dashed curves give comparative black body results at $\lambda = 550$ nm in (a) and $\lambda = 420$ nm in (b).

For DA and DB white dwarfs, the data of Wickramasinghe (1972) enable monochromatic flux derivatives to be derived. Values at $\lambda = 342$ nm and $\lambda = 541$ nm have been obtained with u and y observations in mind. They are listed in Table I under the GD358 and ZZ Ceti regimes. It is not necessary to consider here the relative merits of the more detailed white dwarf atmospheres of Koester *et al.* (1979), Koester (1980), and Wesemael (1981) in view of the paucity of relevant observational data (Section 5.6) and the need for limb-darkening information as outlined in the following subsection.

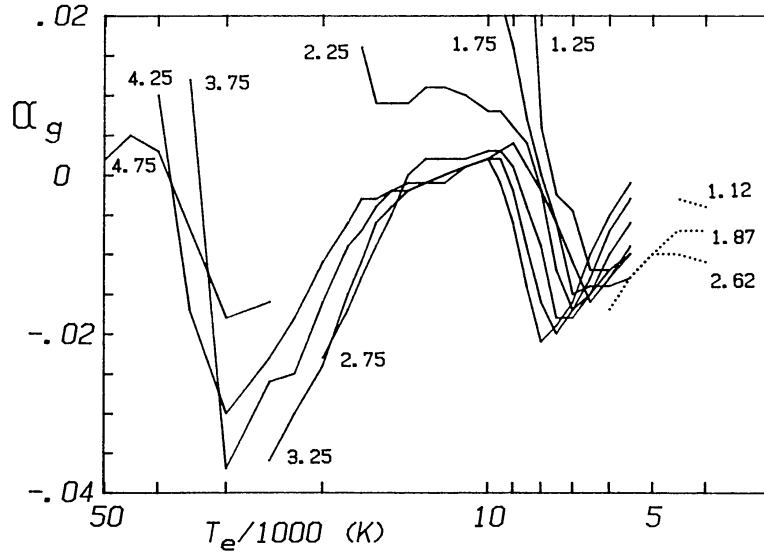


Fig. 3. Dependence of the flux derivative $\alpha_g(\lambda)$ on $(T_e, \log g)$ for solar abundance atmospheres. V band data are shown, and similar conventions to Figure 2 apply.

TABLE I
Some adopted test regime parameters

Regime	$(T_e, \log g)$	Q (d)	λ	$\alpha_T(\lambda)$	$\alpha_g(\lambda)$
β Cephei	(25 000, 4)	0.027	150 nm	3.85	+ 0.002
			U	2.60	- 0.038
			V	1.85	- 0.024
53 Persei	(16 000, 4)	0.50 to 0.5	U	2.60	- 0.028
			V	1.60	+ 0.000
BD - 9°4395	(22 500, 2.5)	0.15, 0.5			
δ Scuti	(8000, 4)	0.03	B	5.00	- 0.043
			V	3.75	- 0.018
Rapid Ap	(8000, 4)	0.0015			
R CrB	(7000, 0.5)	0.06			
Cepheid	(6000, 2)	0.04	B	6.45	- 0.012
			V	4.90	- 0.014
Solar	(6000, 4.5)	0.0035	B	5.90	- 0.009
			V	4.60	- 0.004
Mira	(3000, 0)	0.05	B	13	+ 0.06
			V	10	+ 0.00
PG1159-035	(100 000, 8)	5			
GD358	(25 000, 8)	5	342 nm	0.9	+ 0.005
			541 nm	0.7	+ 0.004
ZZ Ceti	(12 000, 8)	5	342 nm	3.3	+ 0.02
			541 nm	3.0	- 0.03

3.2. LIMB-DARKENING

Wade and Rucinski (1985) provide access to reliable limb-darkening data over the full range of the Kurucz (1979) models. If the normalized limb-darkening law $h_\lambda(\mu)$ is expressed in the quadratic approximation as

$$h_\lambda(\mu) = \chi_0 + \chi_1\mu + \chi_2\mu^2, \quad (16)$$

then χ_1 and χ_2 may be obtained from the tabulated quadratic limb-darkening coefficients of Wade and Rucinski, while $\chi_0 = (2 - 2\chi_1/3 - \chi_2/2)$. Substituting (16) into (7) and integrating gives

$$b_{l\lambda} = c_0 + c_1\chi_1 + c_2\chi_2, \quad (17)$$

TABLE II
Coefficients in $b_{l\lambda} = c_0 + c_1\chi_1 + c_2\chi_2$ using a quadratic limb-darkening law

l	c_0	c_1	c_2
0	+1	0	0
1	$+\frac{2}{3}$	$+\frac{1}{36}$	$+\frac{1}{30}$
2	$+\frac{1}{4}$	$+\frac{1}{20}$	$+\frac{1}{16}$
3	0	$+\frac{1}{24}$	$+\frac{2}{35}$
4	$-\frac{1}{24}$	$+\frac{1}{72}$	$+\frac{5}{192}$
5	0	$-\frac{1}{192}$	0
6	$+\frac{1}{64}$	$-\frac{1}{192}$	$-\frac{7}{1280}$
7	0	$+\frac{1}{640}$	0

where, for low l values, the c_i coefficients are given in Table II. These coefficients indicate the increasing sensitivity of $b_{l\lambda}$ to the form of the limb-darkening law as l increases. The $b_{l\lambda}$ generated by (17) using Wade and Rucinski data are monochromatic values, but the wavelengths selected by these authors are chosen to be close to the effective wavelengths of some common filter bands.

In Figure 4 the strong dependence of $b_{l\lambda}$ on the value of $l < 4$ is illustrated using solar abundances and $\log g = 4$. Wavelength and $\log g$ dependence are then displayed in more detail in Figure 5, with full $(T_e, \log g)$ dependence being shown for $\lambda = 547$ nm. Values for $b_{l\lambda}$, together with temperature and gravity derivatives of it, were estimated from graphs similar to Figure 5 for all normal-abundance variable star test regimes considered here. Monochromatic values at 343, 421, and 547 nm were used to relate to U , B , and V data. An average of monochromatic values at 136 and 181 nm was applied to the case of 150 nm β Cephei data. The accuracy of the generated $b_{l\lambda}$ values is difficult to assess, but is likely better than 1% for $l \leq 2$, degenerating somewhat at cooler temperatures and in cases where line-blocking is appreciable in broadband filters. Particular uncertainty again applies to the Mira case because of large temperature extrapolation. The accuracy of $b_{l\lambda}$ derivatives is probably no better than 30% in most instances.

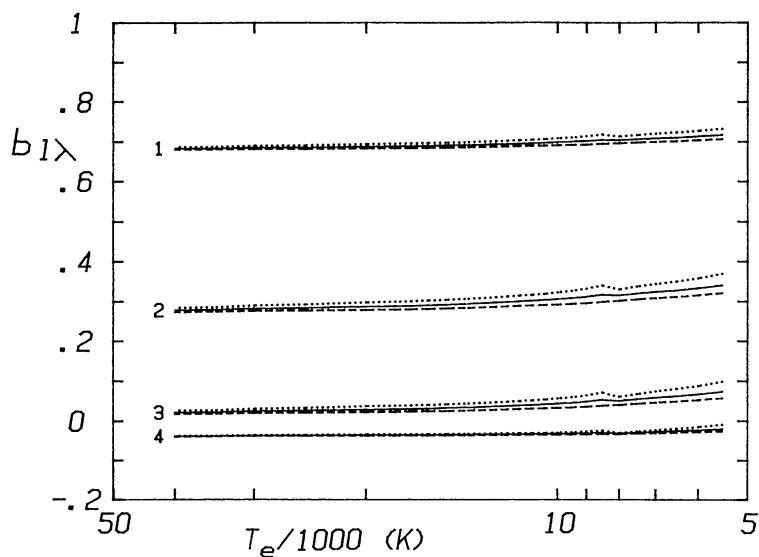


Fig. 4. Dependence of the weighted limb-darkening integral $b_{l\lambda}$ on l . The l values are marked and $b_{0\lambda} = 1$. Data are derived from Wade and Rucinski (1985) using quadratic limb-darkening coefficients, solar abundances and $\log g = 4$. The relative size of the wavelength dependence is illustrated using dotted lines for $\lambda = 421$ nm, full lines for $\lambda = 547$ nm, and dashed lines for $\lambda = 697$ nm.

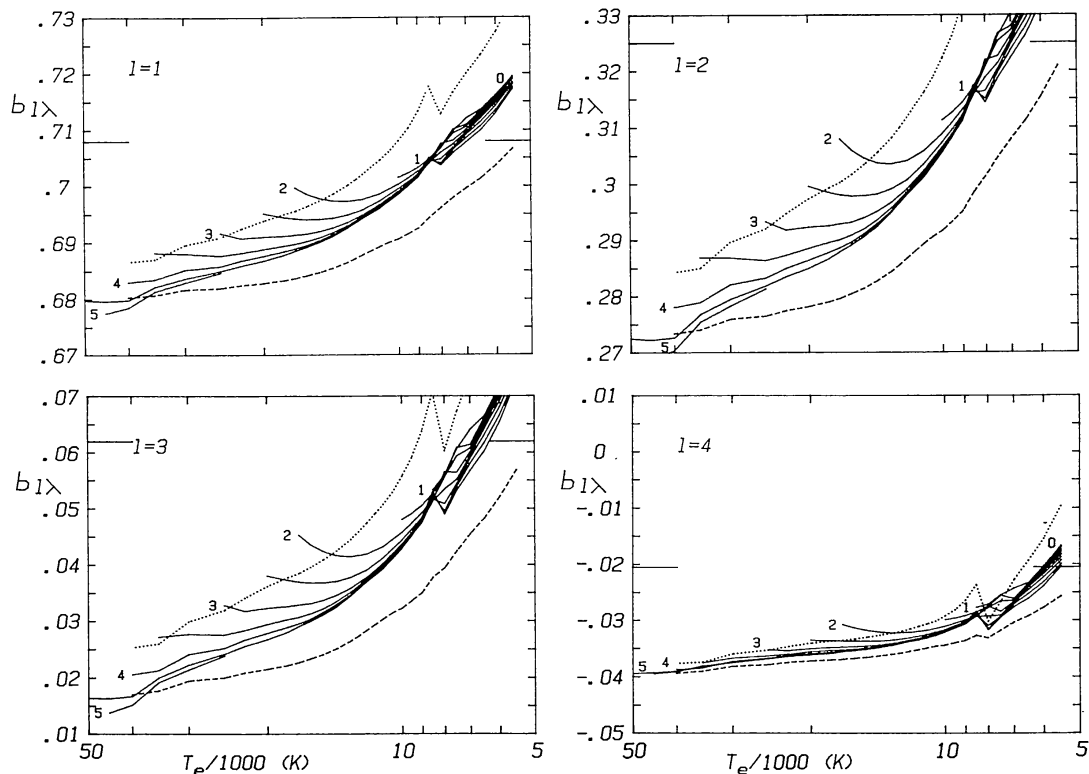


Fig. 5. Details of the $(T_e, \log g)$ dependence of the weighted limb-darkening integral $b_{l\lambda}$. Full lines with $\log g$ values marked are for $\lambda = 547$ nm, thus approximating V band data. For reference, the $\log g = 4$ values of $\lambda = 421$ nm (dotted) and $\lambda = 697$ nm (dashed) are shown. Short horizontal bars indicate, for each l , the corresponding $b_{l\lambda}$ value which would be given by the linear Eddington limb-darkening law.

For the DA and DB white dwarf regimes, a less reliable approach to obtaining $b_{l\lambda}$ is necessitated by the lack of published limb-darkening data. Here the procedure outlined by Robinson *et al.* (1982) was followed. In this procedure the $T(\tau)$ relations given by Wickramasinghe (1972) at $\lambda = 515$ nm were used to generate approximate $T(\tau)$ relations at other wavelengths of interest. This was done by assuming the optical depth scale of these wavelengths differs from that at 515 nm by a constant scale factor which may be determined. From these $T(\tau)$ relations, linear limb-darkening laws were derived and the corresponding $b_{l\lambda}$ values formed. Values at $\lambda = 342$ nm and $\lambda = 541$ nm were thus obtained for approximate application to u and y filter data. Crude estimates of $b_{l\lambda}$ derivatives at these same wavelengths were obtained by repeating the calculations for atmospheres with $(T_e, \log g)$ values near those of the selected white dwarf test regimes.

3.3. PULSATION CONSTANTS

Pulsation constants listed in Table I are representative values for different test regimes and derive from a number of sources. In the β Cephei (Shobbrook, 1985), δ Scuti (Breger and Bregman, 1975), short-period Cepheid (Cox, 1979), Mira (Wood, 1982), and R CrB (Saio and Wheeler, 1983) cases, the listed values are close to expected radial fundamental or first overtone values. In the ZZ Ceti, GD358, and PG1159-035 stars, the periods given in McGraw (1979), Winget *et al.* (1982, 1983) and Starrfield *et al.* (1984) suggest 500 s is a reasonable mid-range value for these groups. Combining this with an assumed $M_* \sim 0.6 M_\odot$ and $R_* \sim 0.01 R_\odot$ yields $Q \sim 5$ d as a representative value, with a range of about a factor of two possible on either side of this. This Q is much greater than the radial fundamental value.

The Q values listed in Table I for the solar and rapidly oscillating Ap star examples are much smaller than the radial fundamental value. In the solar case the listed Q comes from an assumed 300 s period. In the Ap star case it comes from an adopted mid-range period of 500 s (Kurtz, 1982b) combined with values $M_* \sim 2 M_\odot$ and $R_* \sim 3 R_\odot$, which are based on an assumed similarity to δ Scuti conditions.

The two remaining examples in Table I present at least some modes with Q values exceeding radial fundamental ones. These are the 53 Persei variables, for which data (Buta and Smith, 1979; Smith, 1978, 1980b; Smith and McCall, 1978) give Q values ranging from ~ 0.05 d to ~ 0.5 d, and the extreme helium star BD - 9°4395, for which Q values of ~ 0.15 d and ~ 0.5 d apply to the two observed periods (Jeffery *et al.*, 1985).

3.4. OTHER PARAMETERS

Elementary modelling predicts p^* values ranging from $p^* \sim 1$ for hot stars to $p^* \sim 1.5$ to 2 for cool stars (Mihalas, 1970; Gray, 1976). Here values of p^* were estimated directly for each test regime using the defining equation (10) in conjunction with the solar abundance models of Bell *et al.* (1976) and Kurucz (1979), and the white dwarf models of Wickramasinghe (1972). Values all lay in the range $p^* = 1.2$ to 1.6.

As regards the Γ_2 parameter, a value of $\frac{5}{3}$ was adopted for all test regimes except the hot low-gravity ones relating to the β Cephei and BD-9°4395 examples. For these a value of $\frac{4}{3}$ was assumed.

The (R, ψ_T) values are extremely uncertain. Discussion of radially pulsating β Cephei, δ Scuti, and Cepheid examples by Stamford and Watson (1981) indicates that R values as small as 0.25 are seen. Similar values are likely in low l nonradial pulsations. Hence, a value of $R = 0.5$ is adopted as a rough guide in Section 4, and in Section 5 values of 1 and 0.25 are assumed to provide an indication of the range of R . Different ψ_T apply to different regimes. Values of $\psi_T \sim \pi$ are applicable in the β Cephei case (Saio and Cox, 1980), and values $90^\circ < \psi_T < 140^\circ$ apply to the classical Cepheids and δ Scuti stars (Balona and Stobie, 1979b, 1980a).

4. Components of Flux Changes in Example Regimes

The relative importance of the terms T_i , given by Equations (2) to (6), in contributing to predicted flux change is readily visualized as a function of l if one defines

$$\mathbf{R}_i(\text{light}) = |T_i|/(\sum |T_i|), \quad (18)$$

and plots these $\mathbf{R}_i$ in histogram form. This was done for visual wavelengths in various test regimes and the resulting histograms appear as the left-hand panels in Figures 6 and 7. The adopted model parameters follow from Section 3, with an $R = 0.5$ assumed.

The relative importance of the component terms in the predicted colour change may be similarly visualized. Values ΔT_i are formed, being the differences between corresponding T_i values for the two chosen filter bands. Then one derives

$$\mathbf{R}_i(\text{colour}) = |\Delta T_i|/(\sum |\Delta T_i|). \quad (19)$$

Histograms of these values appear as the right-hand panels in Figures 6 and 7.

Figure 6 groups those test regimes, of assumed normal solar abundances, which have Q values close to radial fundamental or first overtone ones. The temperature term T_1 (vertical hatching), is dominant at low l but becomes less important, in both visual light and colour, by $l = 4$. In the visual light, the geometry term T_3 (45° hatching), is a significant contributor in both $l = 0$ and $l = 2$ cases, and becomes dominant by $l = 4$. It gives no contribution for $l = 1$. In the colour histograms, $T_3 = 0$ for both $l = 0$ and $l = 1$ and remains quite small for $l = 2$, but grows thereafter.

It is just this form of l dependence for T_1 and T_3 which Stamford and Watson (1981) argued is the source of $l = 0, 1, 2$ mode discrimination feasibility in β Cephei systems when colour to light amplitude ratios are considered, and in δ Scuti systems when phase differences between colour and light changes are examined. Basically, for $l = 0, 1, 2$, the colour changes remain locked to the T_1 term, at least to the extent that T_2 , T_4 , and T_5 are small. The visual light changes, however, contain the effect of both the temperature and geometry terms, and as (3) shows, the geometry term changes sign as l varies from 0 to 2. The most favourable l discrimination situation requires a choice of filter bands where $(\alpha_T(\lambda_1) - \alpha_T(\lambda_2))$ is large and, at the same time, one of the $\alpha_T(\lambda)$ values is small. This permits accurate observational and model colours while, in one filter band, the geometry term still remains a very significant contributor. The first three test regimes

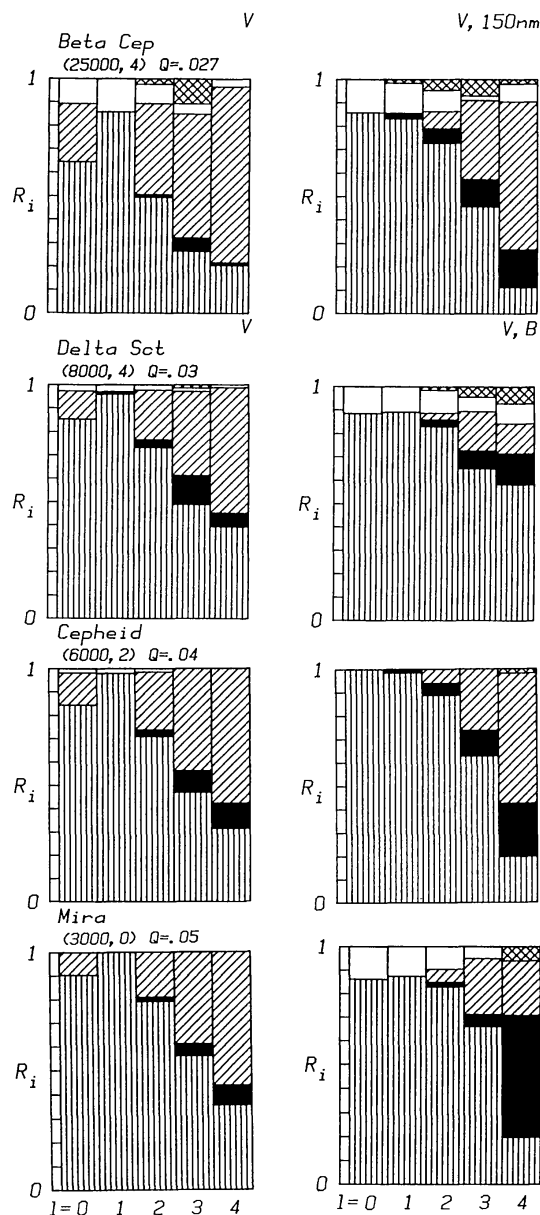


Fig. 6. The relative importance, as a function of l , of the terms T_i in the predicted flux changes for four test regimes which have Q values close to radial fundamental or first overtone ones. Left-hand histograms apply to flux changes in the V band and have R_i defined by (18). Right-hand histograms, which apply to colour changes in (150 nm, V) for the β Cephei case, and to colour changes in (B , V) for the remaining cases, have their R_i defined by (19). The histogram coding is such that vertical hatching applies to T_1 , solid hatching to T_2 , 45° hatching to T_3 , no hatching to T_4 , and cross hatching to T_5 . A value of $R = 0.5$ is assumed.

in Figure 6 are favourable in these regards, but the Mira example involves relatively close and inaccurate $\alpha_T(\lambda)$ values.

While the T_1 and T_3 terms are the most important ones in the Figure 6 histograms, it is clear that pressure effects, T_4 (no hatching), and variable limb-darkening effects, T_2 (solid hatching) and T_5 (cross hatching), cannot always be ignored. Because of the

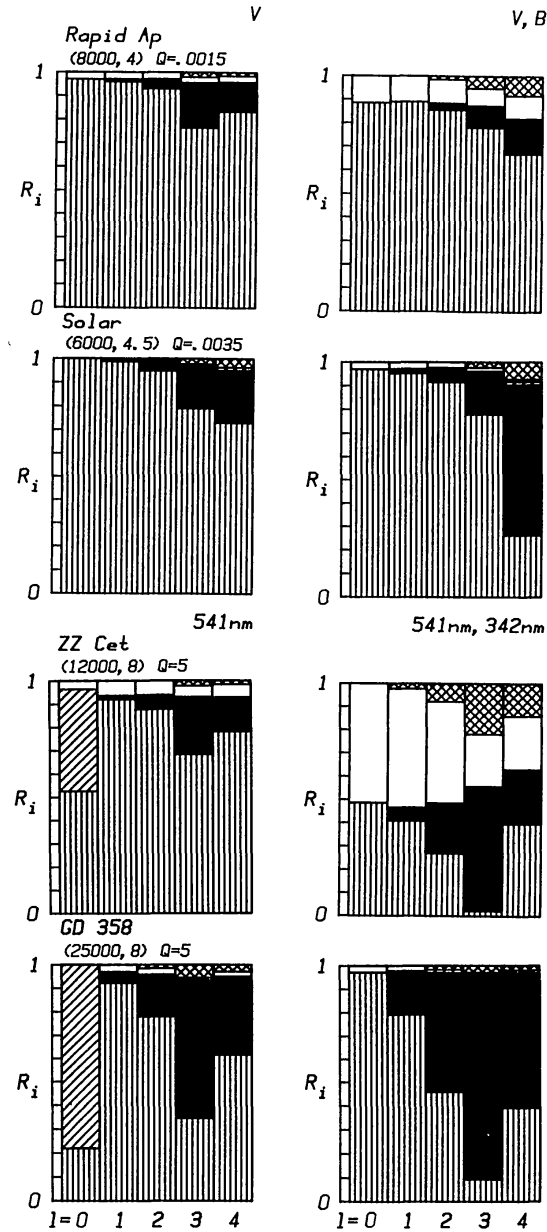


Fig. 7. Similar to Figure 6, but for some test regimes where the Q values are very different from radial fundamental ones. In the rapidly oscillating Ap and solar cases the models are for V and (B, V) . In the white dwarf examples the models are for 541 nm and (342 nm, 541 nm).

low reliability of the relevant parameters, the accuracy of these terms is low, but the T_2 and T_5 terms are notably present if $l > 2$ and, at least in the T_2 case, may get to be quite significant. If the chosen R value in Figure 6 is inappropriate, then the balance between the temperature-related terms T_1 and T_2 and the remaining terms is altered. If, for example, $R < 0.5$ then geometry, pressure, and T_5 terms grow in importance. Finally, it should be noted that if $T_2 = T_5 = 0$ then the histograms are totally independent of $b_{l\lambda}$ and errors associated with it. Hence, cases where T_2 and T_5 are small are also a guide to situations where errors in $b_{l\lambda}$ values are not important.

Figure 7 brings together test regimes which have Q values markedly different from radial fundamental ones. No stellar atmosphere parameters based on realistic abundance values are available for the rapidly oscillating Ap star example. The Figure 7 histograms for this example employ the δ Scuti atmospheric parameters. Thus comparing this Ap star case with the δ Scuti histograms in Figure 6 simply serves to demonstrate the effect of reducing Q from 0.03 d to 0.0015 d. The principal difference between Figure 7 and Figure 6 is that the Figure 7 histograms have a totally negligible contribution from the geometry term T_3 , except in the $l = 0$ case of the large Q white dwarf regimes. With this single exception, the temperature term T_1 , and the related limb-darkening temperature derivative T_2 , completely dominate the predicted visual light changes. The T_1 dominance was noted in the ZZ Ceti case by Robinson *et al.* (1982). When the geometry term and other terms with the same phase are negligible, the rationale which led to a favourable l discrimination situation in the Figure 6 examples is no longer valid. Only in the white dwarf cases does a distinction between $l = 0$ and $l > 0$ remain possible in principle. However, even for low l , the components of the colours in the white dwarf models are unreliable, because they are based on close $\alpha_T(\lambda)$ values.

The fundamental nature of the difference between Figures 6 and 7, as regards the significance of the geometry term, is readily seen by considering the simplest possible model of nonradial flux changes. This simplest model ignores T_2 , T_4 , and T_5 effects, and considers only the predicted bolometric flux change in terms of the interplay between the temperature and geometry terms T_1 and T_3 alone. In this situation Equation (18) gives the relative importance of the geometry term as

$$\mathbf{R}_3 = |T_3|/(|T_1| + |T_3|) = \frac{|(2 + l)(1 - l)|}{|4R(1 - \Gamma_2^{-1})C| + |(2 + l)(1 - l)|}. \quad (20)$$

Equation (20) employs (2), (3), and (12) with $\alpha_T(\lambda) = 4$. Since (11) expresses C in terms of Q and l , $\mathbf{R}_3$ is determined as a function of Q for each l , with only $R(1 - \Gamma_2^{-1})$ as a free parameter. Figure 8 shows these functions of Q , for $l = 0$ to $l = 4$, assuming $R = 1$ and $\Gamma_2 = \frac{5}{3}$.

In Figure 8, the $l = 0$ value of $\mathbf{R}_3$ increases monotonically with increasing Q up to the limit set by the parameter $R(1 - \Gamma_2^{-1})$. The $l = 1$ case has $\mathbf{R}_3 = 0$ since there is no geometry term. For $l > 1$, however, $\mathbf{R}_3$ rises to a maximum at those specific Q values where the temperature term is zero. These values are the roots of the quadratic (11) with $C = 0$. The roots are not greatly different in value from the Q values of radial fundamental oscillations in accurate stellar interior models. Thus, as illustrated in the top panel of Figure 8, those test regimes which have Q values close to radial fundamental ones will have significant geometry terms for $l > 1$. Those with Q values markedly different from radial fundamental ones will have insignificant geometry terms for $l > 1$. The 53 Persei regime and the BD - 9°4395 example have Q values in the intermediate range between these extremes. These basic conclusions remain valid even if the full nonradial flux change expression (1) is used to evaluate $\mathbf{R}_3$. Inclusion of all terms T_1 to T_5 alters the form of $\mathbf{R}_3$, but not the Q values for which $\mathbf{R}_3 = 1$.

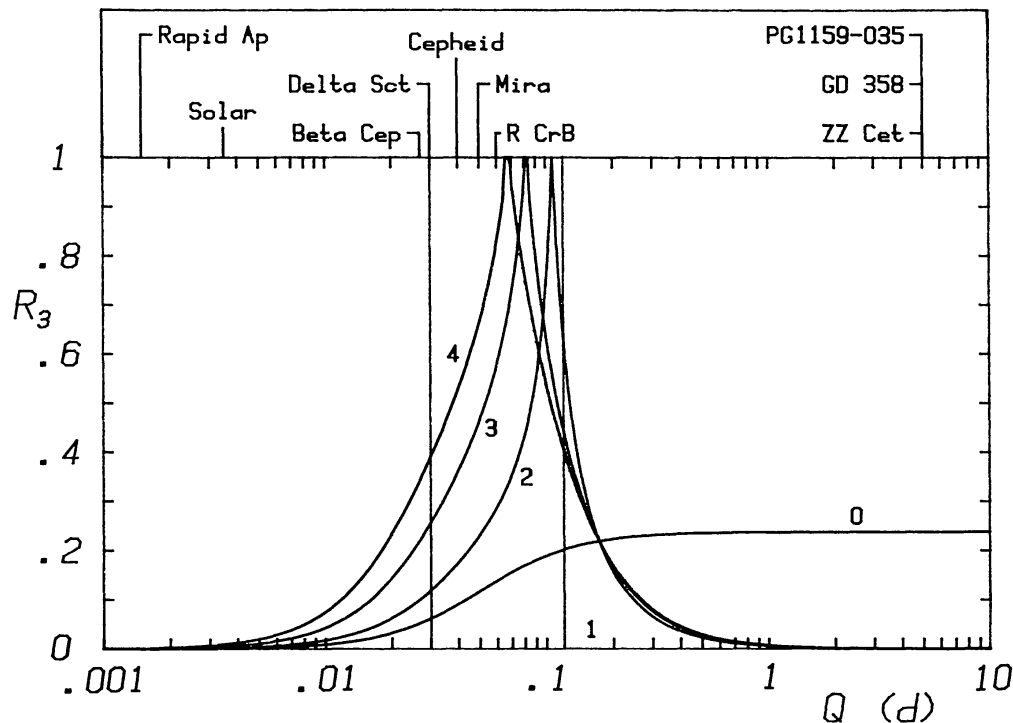


Fig. 8. The relative importance, as a function of Q and l , of the geometry term in the simplest model of bolometric flux change. The value R_3 is defined by (20). An $R = 1$ and a $\Gamma_2 = \frac{5}{3}$ are assumed. Curves are labelled according to l value. $R_3 = 0$ for $l = 1$. Two vertical lines mark the range $0.03 \text{ d} \leq Q \leq 0.12 \text{ d}$. This is noted by Cox (1980) as the extent of radial fundamental Q values in accurate interior models with $\Gamma_2 = \frac{5}{3}$. Test regime Q values are shown in the upper panel.

As already noted, a favourable situation for mode discrimination between $l = 0, 1$, and 2 cases requires the interaction of the geometry term with a well-determined temperature term, plus an accurate colour determination. Hence, a general rule may be stated to the effect that this favourable situation can only occur if the Q value of the pulsation is not grossly different from that of the radial fundamental one. If the Q value is grossly different from the radial fundamental one, then any possibility of mode discrimination would require the interaction of the temperature term with terms other than the geometry term, and such terms are currently poorly determined. It is fortunate that the circumstances here complement those where information on nonradial pulsation is inferred directly from observed Q values. An observed Q value grossly different from the radial fundamental one might lead in itself to the suspected presence of nonradial pulsation, though no mode indices may be inferred.

5. Observational Comparison

A comparison of observational data with the predictions made by (1) is possible as follows. If one writes $\Delta m(\lambda, t) = A_\lambda \cos(\omega t + \phi_\lambda)$ for each of two observed wavelengths or filter bands, then the quantities $(A_{\lambda_1}, \phi_{\lambda_1})$ and $(A_{\lambda_2}, \phi_{\lambda_2})$ are determinable for the

model, except for $[\varepsilon P_{l, |m_l|}(\mu_0)]$. This latter quantity is not required if the comparison with observational data is undertaken on the amplitude ratio versus phase difference plane $(A_{\lambda_2}/A_{\lambda_1}, \phi_{\lambda_2} - \phi_{\lambda_1})$. Alternatively, if a colour amplitude and phase difference $(A_{\text{col}}, \phi_{\text{col}})$ is derived from the model using the two filter bands, the comparison may be made on the $(A_{\text{col}}/A_{\lambda_1}, \phi_{\text{col}} - \phi_{\lambda_1})$ plane. This second form of comparison is mostly employed here because much published observational data are in the $(A_{\text{col}}, \phi_{\text{col}})$ form. The first form of comparison is to be preferred, however, in situations where a small colour amplitude and consequent highly uncertain colour phase disrupt the presentation of an $(A_{\text{col}}/A_{\lambda_1}, \phi_{\text{col}} - \phi_{\lambda_1})$ plot.

In either form of comparison, the model predicts a single point on the comparison plane for each l value. However, the (R, ψ_T) values are very uncertain and may have a substantial effect on the location of points on the comparison plane, even when uncertainties associated with the other parameters have minimal effects. Because of this, loci of constant R and constant ψ_T may be drawn on the comparison plane to select an 'area of interest' for each l value. This selection is based on assumed ranges of (R, ψ_T) , as indicated in Section 3.4. The following subsections discuss the systematic application of this technique to those test regimes for which adequate data exist. This provides a further indication of the limits to the validity of the model and its use as a mode discriminator. The discussion is limited to the $l = 0, 1$, and 2 modes. For higher l , the location of areas of interest on the comparison plane is very volatile, and these areas often overlap those of lower l values.

5.1. β CEPHEI VARIABLES

Figure 9 shows (R, ψ_T) loci on the $(A_{U-V}/A_V, \phi_{U-V} - \phi_V)$ plane for the $l = 0, 1$, and 2 cases of the β Cephei test regime. Panel (d) of this figure then outlines areas of interest selected on the basis of R values lying between 1 and 0.25, and ψ_T values lying within the range $180^\circ \pm 45^\circ$ (Davey, 1970; Saio and Cox, 1980). Panel (d) demonstrates the well established result that quadrupole modes in β Cephei stars are expected to have lower A_{U-V}/A_V ratios than radial modes of pulsation.

Figure 10 demonstrates the sensitivity of the selected areas of interest to some of the other parameters in the nonradial pulsation model. Specifically, the effect of omitting the minor pressure and limb-darkening derivative terms is shown in panel (a), the effect of increasing Q from the 0.027 d value of a first overtone to the 0.037 d value of the radial fundamental is shown in panel (b), and the effect of an error in $\alpha_T(U)$ of 0.1 is shown in panel (c). Radial and quadrupole areas of interest remain relatively stable to such changes.

Observed A_{U-V}/A_V , derived from published data on accepted β Cephei variables, are given in Table III. Error ranges listed in this table are mostly subjective estimates of $\pm 1\sigma$ values, based on original data sources which often do not specify an error. Similarly, most sources do not specify the observed phase difference between colour and light changes, and simply state that these changes are in phase. A reasonable limit seems to be that $|\phi_{U-V} - \phi_V| \leq 30^\circ$ in all cases. In panel (d) of Figure 10, the observed A_{U-V}/A_V are shown superposed on the $l = 0, 1$, and 2 areas of interest. The data are

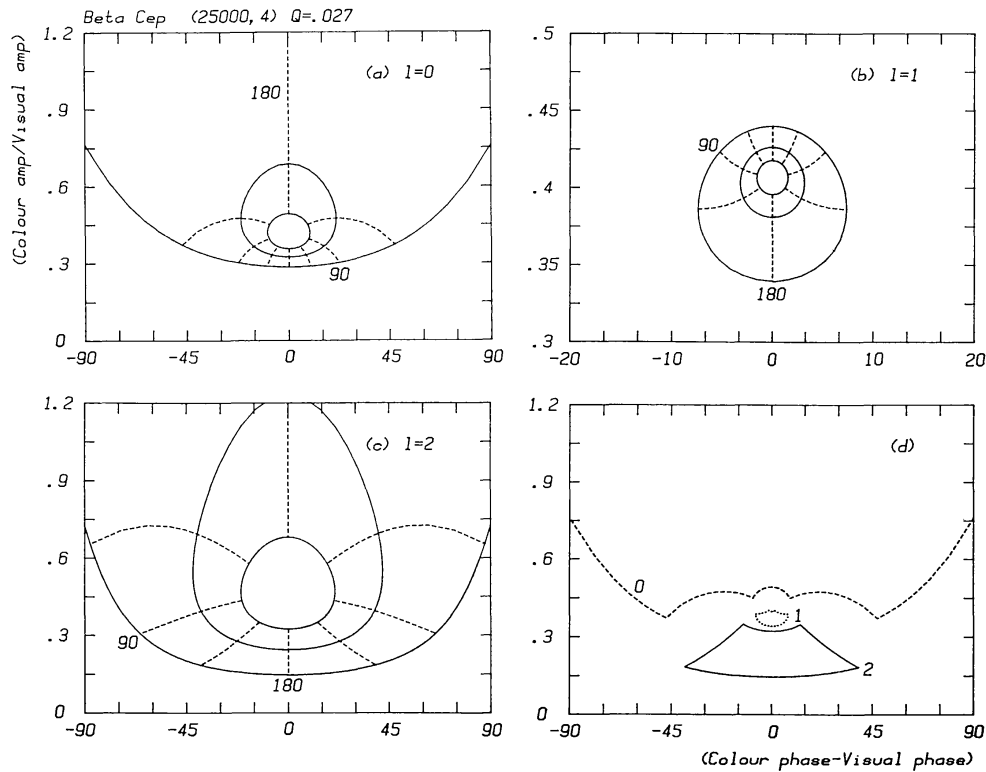


Fig. 9. $(A_{U-V}/A_V, \phi_{U-V} - \phi_V)$ plots of (R, ψ_T) loci for the β Cephei test regime. In panels (a), (b), and (c) the closed curves (full lines) are loci of $R = 1, 0.5$, and 0.25 for $l = 0, 1$, and 2 . The innermost curves are for $R = 1$ and the outermost ones are for $R = 0.25$. In these panels the radial curves (dashed lines) are loci of ψ_T in 45° intervals between 0° and 360° . Labels identify the $\psi_T = 90^\circ$ and $\psi_T = 180^\circ$ curves. Panel (d) outlines areas of interest based on R values of between 1 and 0.25 , and ψ_T values between 135° and 225° . In this panel, areas of interest lie above the dashed curve in the $l = 0$ case and inside the closed curves in the $l = 1$ (dotted) and $l = 2$ (full) cases.

plotted at a nominal value of $(\phi_{U-V} - \phi_V) = 0$, but spread slightly for clarity. The observed A_{U-V}/A_V values clearly infer the presence of both radial and quadrupole modes, as was indicated in the early work of Stamford and Watson (1977). Specific guidelines for β Cephei stars may be derived from Figure 10. These are that $A_{U-V}/A_V \gtrsim 0.45$ favours a radial pulsation interpretation, and that $0.1 \lesssim A_{U-V}/A_V \lesssim 0.35$ favours a quadrupole interpretation. Dipole pulsations occupy a small intermediate region of A_{U-V}/A_V .

Some specific examples of Table III data are shown on the $(A_{U-V}/A_V, \phi_{U-V} - \phi_V)$ plane in Figure 11. Panel (a) displays those cases where independent mode assignments are available through the line profile work of Campos and Smith (1980) and Smith (1980a, 1983). An $l = 0$ assignment was made from line profile studies for γ Peg, δ Cet, β Cep and the large amplitude modes of ν Eri and σ Sco. These all fall in the $l = 0$ region of interest in Figure 11(a). In three cases, ν Eri, σ Sco, and 12 Lac, a nonradial assignment was made from the line profile studies. All three cases fall in the $l = 2$ region of interest in Figure 11(a). Two of these cases had a specific $l = 2$ assignment given to them in the line profile studies. Hence, the line profile studies and the format used here

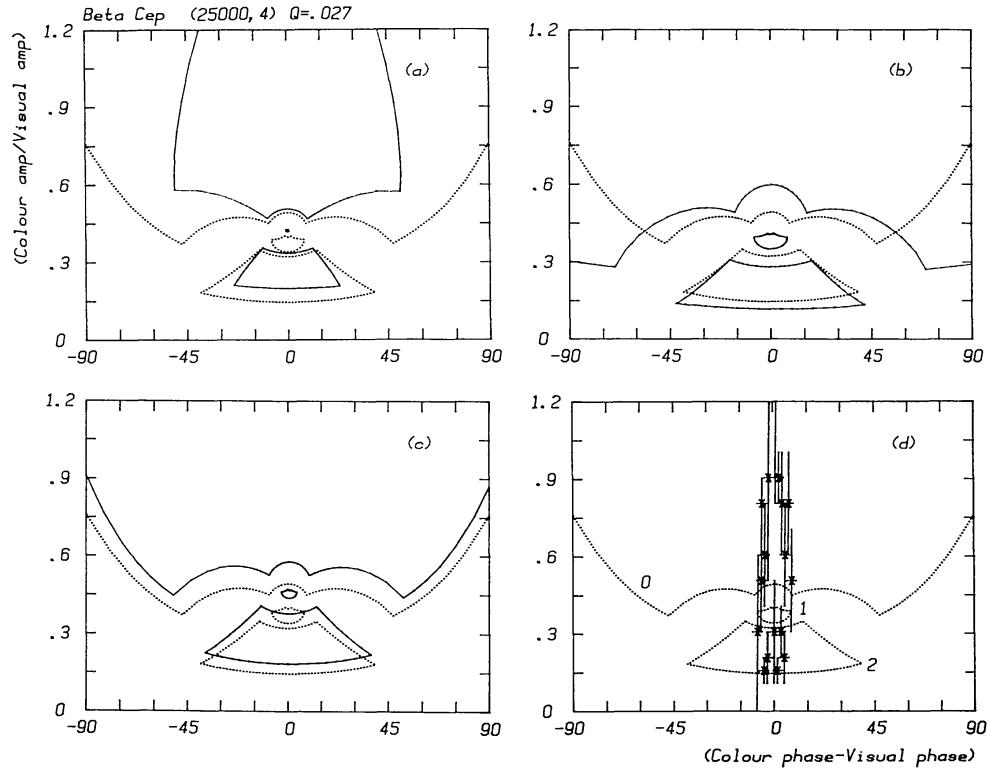


Fig. 10. Parameter sensitivity of areas of interest on the $(A_{U-V}/A_V, \phi_{U-V} - \phi_V)$ plane for the β Cephei test regime. The dotted curves delineate the areas of interest for $l = 0, 1, 2$, as specified for the standard β Cephei test regime in Figure 9(d). In panels (a), (b), and (c), the full curves demonstrate how the boundaries to these areas of interest change if: (a) T_2, T_4 , and T_5 are omitted; (b) Q is increased to 0.037 d; (c) $\alpha_T(U)$ is increased by 0.1. Panel (d) shows the A_{U-V}/A_V data in Table III superposed on the areas of interest for the standard β Cephei test regime. Data points nominally have $(\phi_{U-V} - \phi_V) = 0$ but are spread for clarity.

for the presentation of the $\Delta m(\lambda, t)$ model are fully consistent. Figure 11(b) displays some other systems for which data in Table III clearly favour a radial interpretation (ξ^1 CMa, HR6684, BW Vul) or a nonradial interpretation (β Cru, α Vir).

To the extent that phase information is not relevant in the β Cephei case, one may use the A_{U-V}/A_V guidelines of $\gtrsim 0.45$ for $l = 0$, and ~ 0.1 to ~ 0.35 for $l = 2$, to provide a very simple appraisal of the data in Table III as regards mode discrimination. This is shown in Figure 12.

A more recent example not listed in Table III is the system HD 129929. Waelkens and Rufener (1983) determine three periods between 3.4 and 3.7 hr for this system, suggesting a relation to the β Cephei class. Their observations, on the Geneva system, were assumed transformed to the Johnson system by $A_{U-V}/A_V \sim 0.9 A_{[U-B_2]}/A_{[V]}$. This approximate transform comes from consideration of the temperature dependence of the colours in both the Johnson system and the Geneva system (North and Hauk, 1979). Such a transform suffers from the fact that the observed flux changes in modes

TABLE III
Data for β Cephei variables

Star	HD	$P(\text{hr})$	$A_{150-\nu}/A_V$	$A_{U-\nu}/A_V$	Notes
γ Peg	886	3.64	3.0 ± 0.8	0.9 ± 0.4	1, 2, 3, 4
δ Cet	16582	3.87	2.4 ± 0.3	0.8 ± 0.1	1, 2, 3, 4
ν Eri	29248	4.16		0.8 ± 0.2	3, 5
		4.26		0.3 ± 0.3	3, 6
β CMa	44743	6.00	3.5 ± 2.5		3, 7
		6.03	0.7 ± 0.5		3, 7
ξ^1 CMa	46328	5.03	3.6 ± 0.4	0.9 ± 0.1	1, 3, 7, 8
15 CMa	50707	4.43	1.6 ± 0.5	0.3 ± 0.2	3, 8, 9
HR 2928	61068	3.87	2.1 ± 0.5		1, 8, 10
β Cru	111123	3.85		0.15 ± 0.05	1, 3
α Vir	116658	4.17		0.2 ± 0.1	3, 11
τ^1 Lup	126341	4.26		0.3 ± 0.1	1, 3
α Lup	129056	6.24	2.2 ± 0.6	0.6 ± 0.2	1, 3, 12, 13
σ Sco	147165	5.75		0.15 ± 0.05	14
		5.92	7.5 ± 1.0	1.5 ± 0.05	4, 7, 14
θ Oph	157056	3.37	1.9 ± 0.5	0.6 ± 0.2	2, 3, 15
λ Sco	158926	5.13	2.2 ± 1.2	0.5 ± 0.2	1, 3, 7
HR 6684	163472	3.36	3.6 ± 1.7	1.3 ± 0.5	1, 7, 16
BW Vul	199140	4.82	2.7 ± 0.4	0.8 ± 0.2	1, 3, 13, 17
β Cep	205021	4.57	2.9 ± 0.3	0.5 ± 0.2	1, 3, 4, 7, 18, 19
12 Lac	214993	4.63	0.3 ± 0.3	0.2 ± 0.1	3, 7, 20
16 Lac	216916	4.06	1.0 ± 0.3		3, 7, 21

Notes to Table III

1. Singly periodic or at least one mode dominant.
2. Lesh (1976) is source of A_{150} data.
3. Stamford and Watson (1977) list references to A_U and A_V data.
4. Line profile studies of Campos and Smith (1980) imply radial pulsation.
5. Line profile studies of Smith (1983) imply this mode is radial.
6. Kubiak (1980) identifies this mode as a close triplet and line profile studies of Smith (1983) imply $l = 2$ for the triplet.
7. Beeckmans and Berger (1977) is source of A_{150} data.
8. Lesh and Wesselius (1979) is source of A_{150} data.
9. Shobbrook (1973) suggests probably singly periodic but of variable amplitude. Perhaps other periods sometimes present.
10. ΔV data estimated from published A_{330} of Lesh and Wesselius (1979).
11. Line profile studies of Smith (1985) imply a variety of nonradial modes present.
12. Lampens and Goosens (1982) is source of A_U data.
13. Lesh (1978) is source of A_{150} data.
14. Jerzykiewicz and Sterken (1984) A_u and A_y data are source of A_U and A_V estimates.
15. Further uncertainty due to the possible multiperiodic nature of this object and limited colour data.
16. Norton and Hansen (1974) is source of A_U and A_V data.
17. Barry *et al.* (1984) is a further source of A_{150} data.
18. Hutchings and Hill (1977) is a further source of A_{150} data.
19. Stebbins and Kron (1954) and Hutchings and Hill (1977) are the source of A_U estimates.
20. Jerzykiewicz (1978) identifies a close triplet of modes and a fourth presumably radial mode. Line profile studies of Smith (1980) imply $l = 2$ for the triplet, which contains the dominant mode for which data estimates are presented here.
21. Further uncertainty due to multiple modes and nonsimultaneous A_V and A_{150} data.

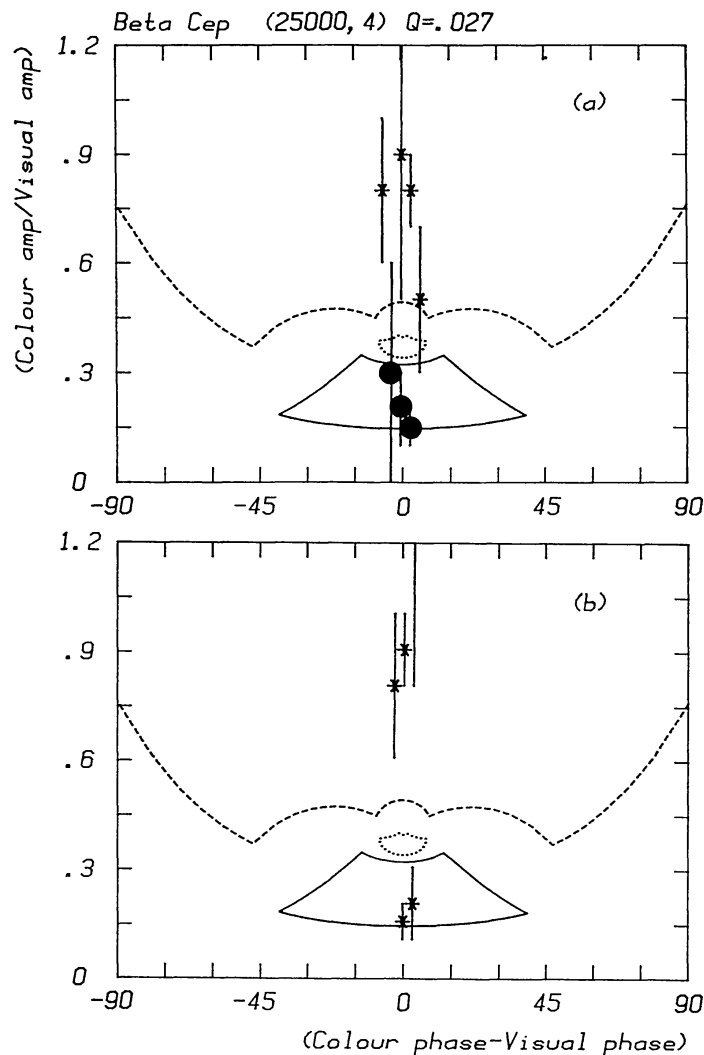


Fig. 11. Selected Table III data compared with the areas of interest on the $(A_{U-V}/A_V, \phi_{U-V} - \phi_V)$ plane for the β Cephei test regime. Dashed, dotted, and full lines correspond, respectively, to $l = 0, 1$, and 2 . In panel (a) the data are for those modes which have been assessed independently by line profile studies as radial pulsations (asterisks) or nonradial pulsations (filled circles). In panel (b) the data are for those systems for which the observed A_{U-V}/A_V , as plotted here, clearly favours a radial interpretation (ξ^1 CMa, HR 6684 [off scale], BW Vul) or a nonradial interpretation (β Cru, α Vir).

may have a different wavelength dependence from that dependence expected for temperature changes only. Bearing this in mind, it appears that the transformed data, plotted on the $(A_{U-V}/A_V, \phi_{U-V} - \phi_V)$ plane in Figure 13, favour dipole or quadrupole modes. This is unusual in that a clear radial mode is normally seen in multiperiodic β Cephei systems.

Finally, it should be recalled that β Cephei satellite ultraviolet data provide a similar but independent framework to that discussed for A_{U-V}/A_V data. Table III lists, as an example, published satellite ultraviolet data, interpolated to 150 nm, in the form A_{150-V}/A_V . Figure 14(d) shows this data superposed on areas of interest for $l = 0, 1$,

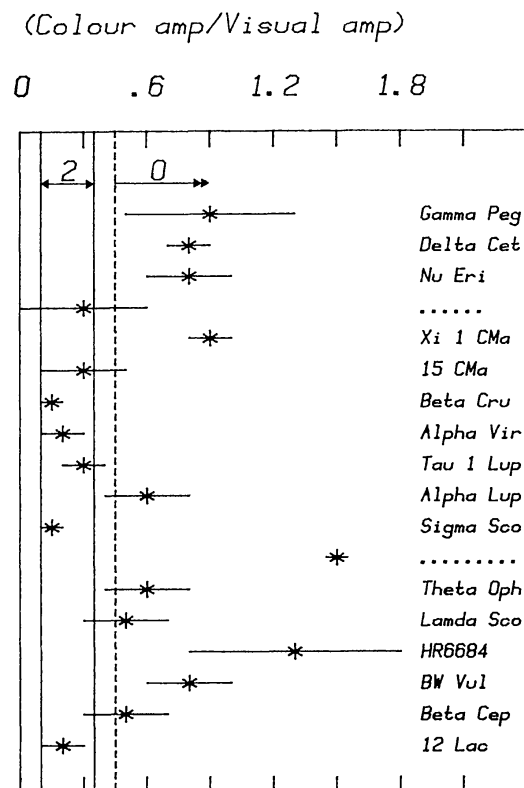


Fig. 12. A_{U-V}/A_V data for β Cephei stars, with radial pulsation ($\gtrsim 0.45$) and quadrupole pulsation (~ 0.1 to ~ 0.35) guidelines shown. Dipole pulsations occupy a small intermediate region.

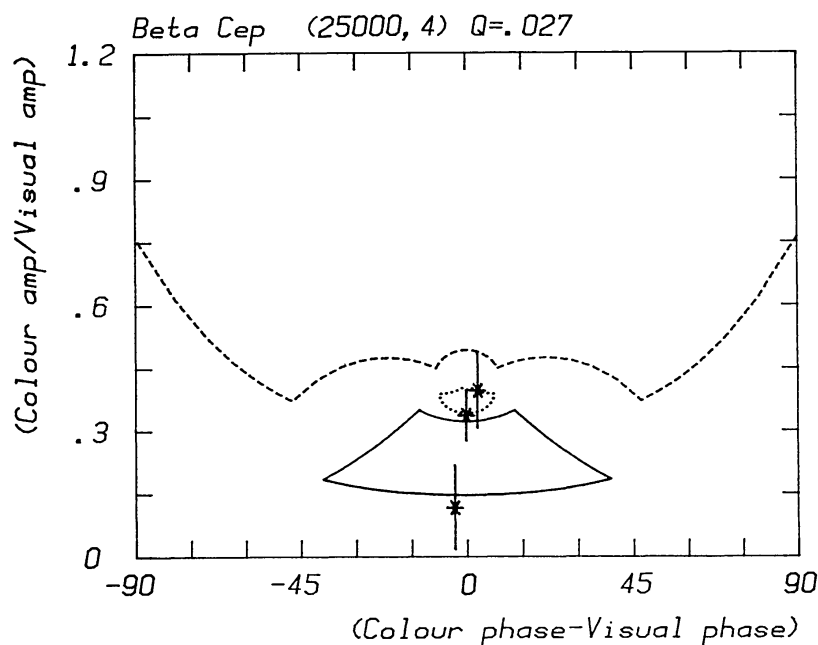


Fig. 13. Data for the system HD 129929 compared with the areas of interest on the $(A_{U-V}/A_V, \phi_{U-V} - \phi_V)$ plane for the β Cephei test regime. Dashed, dotted, and full lines correspond, respectively, to $l = 0, 1$, and 2 . Waelkens and Rufener's (1983) Geneva photometry for the three periods has been converted assuming $A_{U-V}/A_V \sim 0.9 A_{[U-B_2]}/A_{[V]}$.

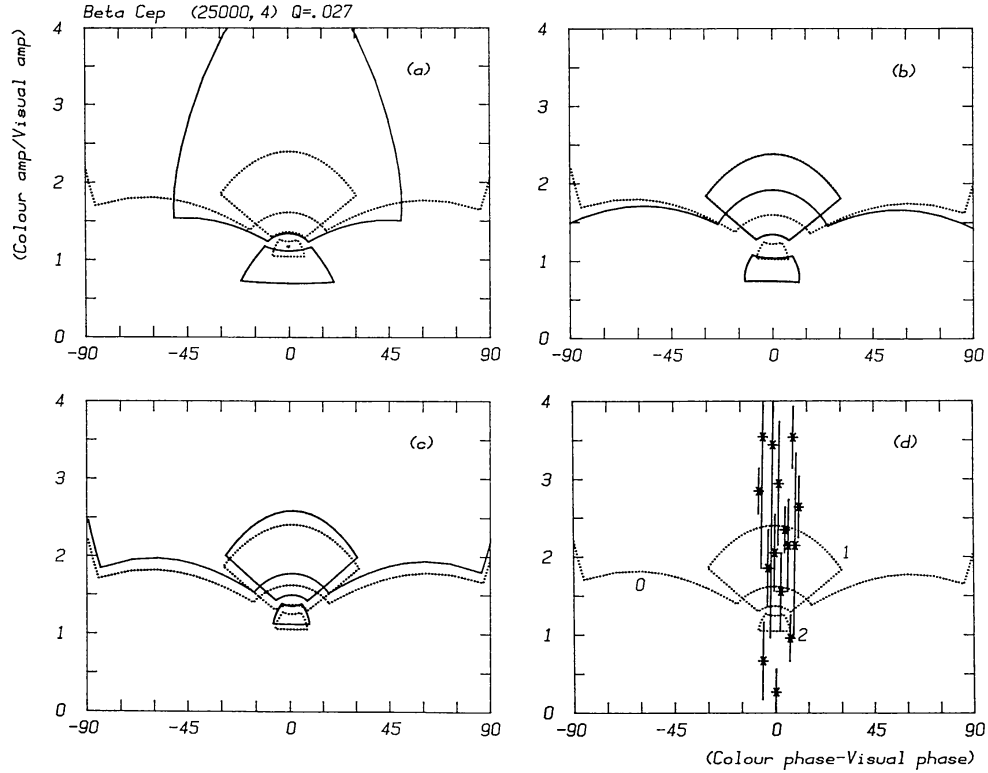


Fig. 14. Parameter sensitivity of areas of interest on the $(A_{150-\nu}/A_V, \phi_{150-\nu} - \phi_V)$ plane for the β Cephei test regime. The format is similar to Figure 10, showing with full curves in panels (a), (b), and (c) the variations that would occur in the standard model boundaries (dotted) if: (a) T_2 , T_4 , and T_5 are omitted; (b) Q is increased to 0.037 d; (c) $\alpha_T(150)$ is increased by 0.2. Panel (d) shows the $A_{150-\nu}/A_V$ data in Table III superposed on the standard model. It is spread, for clarity, about the nominal $(\phi_{150-\nu} - \phi_V) = 0$ value.

and 2 on the $(A_{150-\nu}/A_V, \phi_{150-\nu} - \phi_V)$ plane. The remaining panels of Figure 14 demonstrate the sensitivity of these areas of interest to other parameters in the model, in a similar way that Figure 10 did for the $(A_{U-\nu}/A_V, \phi_{U-\nu} - \phi_V)$ case.

5.2. 53 PERSEI VARIABLES

These variables are a loose grouping of early to mid-B star variables suggested as being g -mode nonradial pulsators (Smith, 1981). Studies of these systems have principally related to line profile variations, but for the prototype system 53 Per, Smith *et al.* (1984) have published v and y photometry on the 1.7 d and 2.2 d periods detected. Since $(\alpha_T(v) - \alpha_T(y))$ is small, the nonradial model is unfortunately not a reliable interpreter of these data, though, even bearing this in mind, the observed $(\phi_v - \phi_y)$ values are rather larger than the model appears to permit.

More recently, however, Waelkens and Rufener (1985) have obtained Geneva $[U]$ and $[B]$ band photometry of seven mid-B star variables which they suspect as being related to the 53 Persei class. The larger value of $(\alpha_T([U]) - \alpha_T([B]))$ presents a more reliable basis on which to attempt to apply the nonradial model (1). The seven stars have

spectral types B4 to B8, luminosity classes III and IV, and periods in the one to three day range. Because of this, a cool, long-period 53 Persei test regime with $(T_e, \log g, Q) = (16000, 4, 0.5)$ was chosen here. Figure 15 outlines areas of interest for this regime, based on ranges in (R, ψ_T) values which are taken to be the same as those adopted for the β Cephei regime. More extreme non-adiabatic conditions may, in fact, prevail in the longer period 53 Persei variables (Smith, 1980c).

Figure 15 has two panels, which involve the same data, but reflect different attempts to deal with the fact that the Waelkens and Rufener observations are on the Geneva

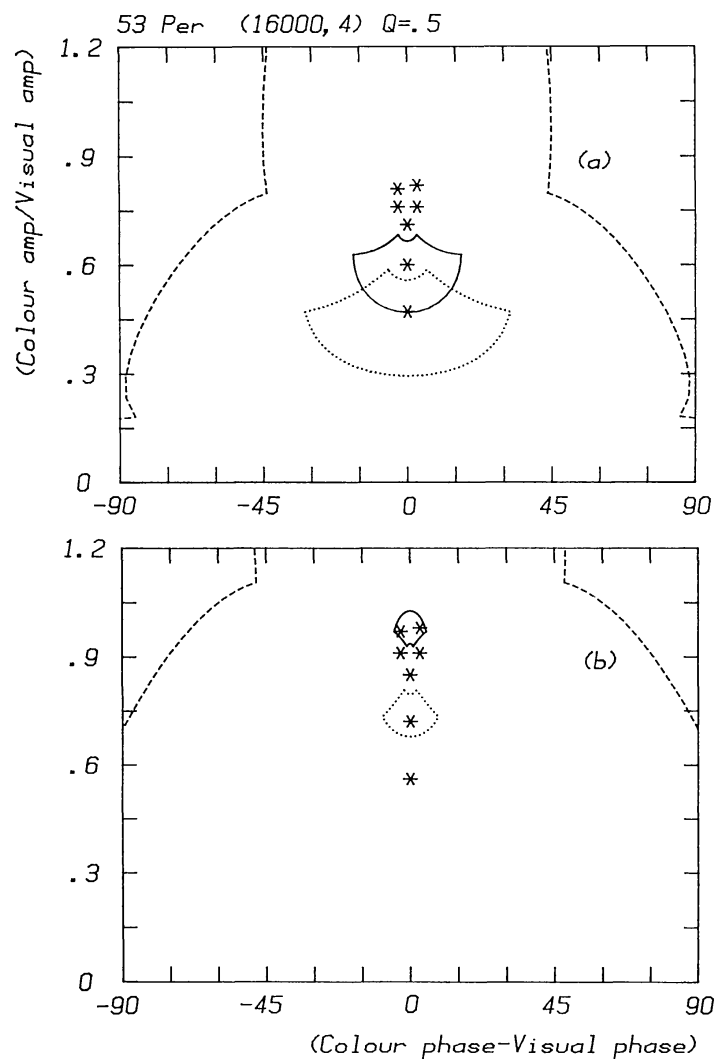


Fig. 15. The mid-B star data of Waelkens and Rufener (1985) compared with areas of interest for a cool, long-period 53 Persei test regime having $(T_e, \log g, Q) = (16000, 4, 0.5)$. These areas of interest are based on R values of 1 to 0.25 and ψ_T values of between 135° and 225° . They lie above and outside the dashed curves for $l=0$, and inside the closed curves for $l=1$ (dotted) and $l=2$ (full). Panel (a) is for $(A_{U-V}/A_V, \phi_{U-V} - \phi_V)$. Panel (b) is a similar plot of the Geneva system $(A_{[U-V]}/A_{[V]}, \phi_{[U-V]} - \phi_{[V]})$. The relevant photometric transforms are discussed in the text.

system rather than on the Johnson system. Panel (a) is an $(A_{U-V}/A_V, \phi_{U-V} - \phi_V)$ plot, with data transformed by $A_{U-V}/A_V \sim 1.0A_{[U-B]}/A_{[V]}$. This transformation comes from the temperature dependence of the Geneva colours of North and Hauk (1979), and suffers from the same limitations discussed in the HD 129929 case of Section 5.1. Panel (b) is an $(A_{[U-V]}/A_{[V]}, \phi_{[U-V]} - \phi_{[V]})$ plot. To generate the required nonradial model for this panel, it was assumed that $\alpha_T([V]) \sim \alpha_T(V)$ and $\alpha_g([V]) \sim \alpha_g(V)$, and then $\alpha_T([U])$ and $\alpha_g([U])$ were derived from North and Hauk colours. Data were transformed using $A_{[U-V]}/A_{[V]} \sim 1.2A_{[U-B]}/A_{[V]}$, with the previously referred to limitations applying. Data in both panels are plotted at a nominal value of zero on the phase axis, since Waelkens and Rufener state that light and colour variations are in phase.

In Figure 15(a) and (b), the observed data lie in or near the dipole and quadrupole areas of interest, and are excluded from the radial area of interest. This provides support for Waelkens and Rufener's (1985) hypothesis of nonradial pulsation in these stars. The result is not sensitive to the particular T_e adopted for the 53 Persei test regime.

As noted by Watson (1987), the differences between the areas of interest here (Figure 15(a)), and those in the corresponding β Cephei test regime (Figure 9(d)), are due to the order of magnitude difference in the Q values between these two cases. Whereas the temperature and geometry terms in (1), T_1 and T_3 , are in phase for $l = 2$ in the small Q β Cephei case, they are out of phase for $l = 2$ in the larger Q case of the 53 Persei test regime. This is simply because of the changed sign of C in (11). The change shifts the quadrupole region of interest from A_{U-V}/A_V values smaller than those of the dipole region of interest in the β Cephei case, to values larger than the dipole region of interest in the 53 Persei test regime. Similarly, the increase of Q raises the permitted lower limit to A_{U-V}/A_V values for the radial case on the phase difference axis, due to the decreasing C value in (11) producing a relatively more important geometric term in the flux change expression (1). In summary, therefore, A_{U-V}/A_V values for long period 53 Persei stars may be larger than those for β Cephei systems, yet still be consistent with the presence of low l nonradial pulsations.

5.3. δ SCUTI VARIABLES

Figure 16 shows (R, ψ_T) locii on the $(A_{B-V}/A_V, \phi_{B-V} - \phi_V)$ plane for the $l = 0, 1$, and 2 cases of the δ Scuti test regime. In this diagram, panel (d) outlines areas of interest selected on the assumption of R values lying between 1 and 0.25, and ψ_T values lying between 90° and 140° (Stellingwerf, 1979; Balona and Stobie, 1980a). Figure 16(d) displays the expected feature (Balona and Stobie, 1979b) of a segregation of areas of interest along the phase difference axis.

The sensitivity of the selected areas of interest to other nonradial model parameters is shown in Figure 17. Figure 17(a) demonstrates the effect of omitting the T_2 , T_4 , and T_5 terms, while Figure 17(b) displays the effect of increasing Q from 0.03 d to 0.04 d. In both cases, the range of phases in the areas of interest is substantially altered, particularly for $l = 0$. Adopting a lower limit to R of less than 0.25 when selecting areas of interest would similarly effect the range of phases. Figure 17(c) demonstrates the

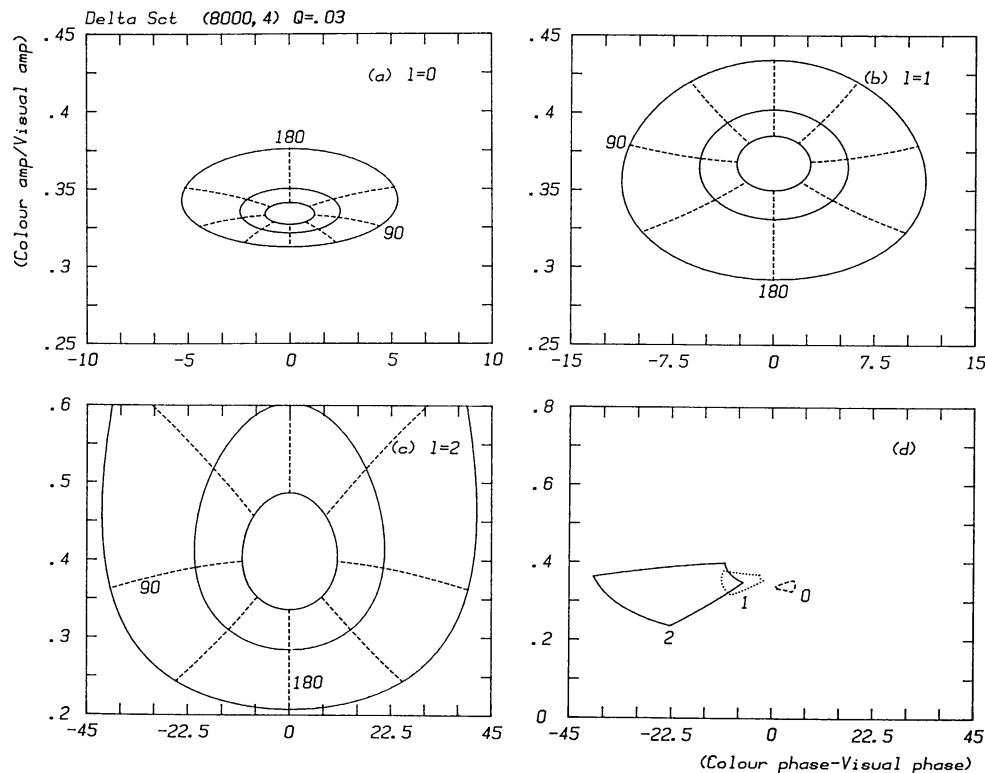


Fig. 16. $(A_{B-V}/A_V, \phi_{B-V} - \phi_V)$ plots of (R, ψ_T) loci for the δ Scuti test regime. The format is similar to Figure 9. Panel (d) outlines areas of interest based on R values of between 1 and 0.25, and ψ_T values of between 90° and 140° . The areas of interest lie inside the closed curves, which are dashed for $l = 0$, dotted for $l = 1$ and full for $l = 2$.

effect of increasing $\alpha_T(B)$ by the substantial amount of 0.5. This principally alters the A_{B-V}/A_V values and has little effect on the range of phases.

Table IV gives values for A_{B-V}/A_V and $(\phi_{B-V} - \phi_V)$ for δ Scuti stars derived from a survey of published data. Except for HD 132209, the frequency identification numbers listed in this table are those cited in the original references. In some instances, the colour amplitudes and phase differences presented in Table IV had to be derived from A_B , A_V , ϕ_B , and ϕ_V data in the original reference. Table IV data are shown in Figure 17(d) superposed on the $l = 0, 1$, and 2 areas of interest for the standard δ Scuti test regime. These data show a general agreement with the areas of interest in the model, particularly as regards the elongation of the data along the phase axis. The latter would be expected if both radial and low l nonradial modes are present in the stars. There are, however, some difficulties. Data for BD + $28^\circ 1494$ have excessively large negative phase differences, placing it off scale in Figure 17(d). Data with positive phase differences have a worryingly large scatter about the $l = 0$ region of interest. This may be due in part to the need to take specific models applicable to individual stars, rather than comparing with a single standard model. It may also indicate the presence of errors in the adopted parameters, more extreme non-adiabatic effects, or other specific effects, such as metallicity in the case of HD 188136.

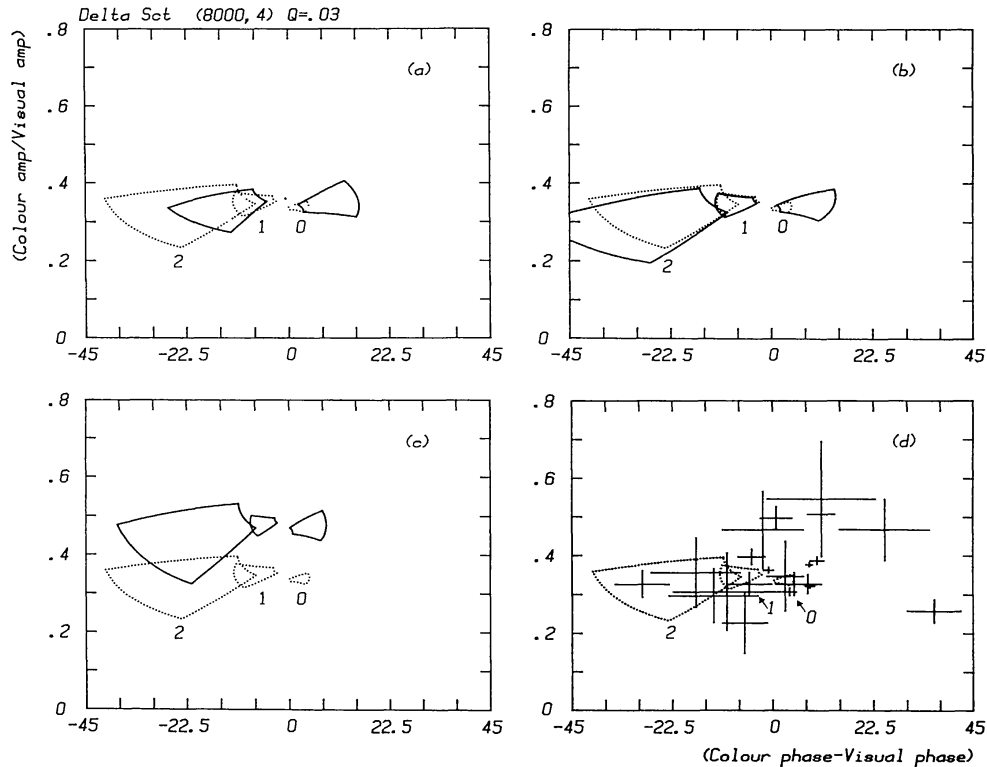


Fig. 17. Parameter sensitivity of areas of interest on the $(A_{B-V}/A_V, \phi_{B-V} - \phi_V)$ plane for the δ Scuti test regime. The format is similar to Figure 10, showing with full curves in panels (a), (b), and (c) the variations that would occur in the standard model boundaries of Figure 16(d) (dotted) if: (a) T_2, T_4 , and T_5 are omitted; (b) Q is increased to 0.04 d; (c) $\alpha_T(B)$ is increased by 0.5. Panel (d) shows the data in Table IV superposed on the standard model.

Certainly it is possible to achieve consistency between observations of some well studied stars and the nonradial model. This is shown in Figures 18 and 19 for the stars δ Sct and 1 Mon, respectively. Panel (a) in these diagrams compares the observational data with the standard δ Scuti test regime, which has areas of interest as in Figure 16(d). Panel (b) in these diagrams compares the same observational data with δ Scuti models having $(T_e, \log g, Q)$ values equal to (7400, 3.5, 0.04) in the δ Sct case, and equal to (7400, 3.5, 0.03) in the 1 Mon case. These are closer to the parameters listed for these stars by Breger and Bregman (1975) and Tsvetkov (1985). In both cases, the panel (b) model value of $\alpha_T(B)$ was increased arbitrarily by 0.3 to achieve better agreement with the data, suggesting the model atmosphere values of this parameter are somewhat in error. A decrease of $\alpha_T(V)$ by ~ 0.2 would produce a similar effect.

The Figure 18 indication of an $l=0$ and $l=2$ mode combination for δ Sct is consistent with the spectral line study of Smith (1982), and the careful work of Balona *et al.* (1981) based on photometry and radial velocities. The 1 Mon plot in Figure 19 is consistent with Smith's (1982) spectral assessment of $l=0$ and $l=1$ modes. Balona and Stobie (1980a) also assess a radial mode and a pair of dipole modes in 1 Mon, using a colour phase study similar to the present approach.

TABLE IV
Data for δ Scuti variables

HD		P (hr)	A_{B-V}/A_V	$(\phi_{B-V} - \phi_V)$ (deg)	Reference
4948		1.32	0.36 ± 0.09	-17 ± 10	Kreidl (1985)
23728	# 1	2.38	0.26 ± 0.03	$+36 \pm 6$	Kurtz (1981b)
	# 2	2.20	0.47 ± 0.08	$+25 \pm 10$	
37819		4.54	0.50 ± 0.03	$+1 \pm 3.5$	Burki and Mayor (1981) ^b
40535	# 1	3.27	0.380 ± 0.005	$+8.3 \pm 0.7$	Balona and Stobie (1980a)
	# 2	3.21	0.37 ± 0.01	-0.7 ± 1	
	# 3	3.32	0.40 ± 0.02	-4.5 ± 3	
69213	# 0	2.68	0.325 ± 0.005	$+8.5 \pm 1$	Balona and Stobie (1980b) ^c
	# 1	2.07	0.31 ± 0.01	$+4 \pm 1.5$	
116994		2.45	0.33 ± 0.03	$+8 \pm 3$	Kurtz (1981d)
132209	# 1	3.79	0.33 ± 0.03	$+5 \pm 4$	Kurtz (1981c)
	# 2	1.95	0.33 ± 0.03	-5 ± 5	
	# 3	3.04	0.30 ± 0.07	-13 ± 10	
	# 4	2.06	0.31 ± 0.10	-10 ± 12	
172748	# 0	4.65	0.39 ± 0.01	$+10 \pm 1.5$	Balona <i>et al.</i> (1981)
	# 1	4.48	0.33 ± 0.04	-29 ± 6	
188136	# 1	3.00	0.51 ± 0.03	$+11 \pm 3$	Kurtz (1980b)
	# 2	1.91	0.55 ± 0.15	$+11 \pm 12$	
	# 3	2.96	0.47 ± 0.10	-2 ± 9	
195961		2.74	0.35 ± 0.09	$+3 \pm 4$	Kurtz (1981a) ^d
208664		1.13	0.23 ± 0.08	-6 ± 5	Kurtz (1982a)
BD + 28° 1494	# 1	1.55	0.46 ± 0.07	-52 ± 8	Broglia and Conconi (1985)
	# 2	1.51	0.36 ± 0.08	-59 ± 13	
	# 3	1.36	0.53 ± 0.08	-54 ± 9	
	# 4	1.32	0.35 ± 0.12	-129 ± 20	

Notes to Table IV

^a Errors are 1σ values.

^b Observations of HD 37819 were in the Geneva system. A factor of 0.8 has been applied to the colour amplitude to estimate A_{B-V} .

^c The AI Vel data are from the fundamental Fourier component of each mode.

^d Observations of ρ Pav were in the b and y filters. A factor of 1.3 has been applied to the colour amplitude to estimate A_{B-V} .

To the extent that amplitude information is less important than phase information as regards l discrimination in the δ Scuti regime, some approximate $(\phi_{B-V} - \phi_V)$ guidelines may be adopted. Values of $\sim 0^\circ$ divide the $l = 0$ and $l = 1$ cases, though some overlap appears possible if the full range of feasible δ Scuti (T_e , $\log g$, Q) models is considered. Values of $(\phi_{B-V} - \phi_V)$ out to $\sim +20^\circ$ are possible for $l = 0$ without reducing the $R = 0.25$ limit. Dipole values extend as far as $\sim -10^\circ$, and quadrupole values range from $\sim -5^\circ$ to $\sim -50^\circ$ without reducing this R limit. These guidelines permit some assertions to be made in support of other mode identification techniques, even if the situation is not as clear-cut as it was in the case of amplitude ratio data for Section 5.1. As an example, Figure 20 supports the claim that main and secondary variations in AI Vel are both radial modes, presumably fundamental and first overtone.

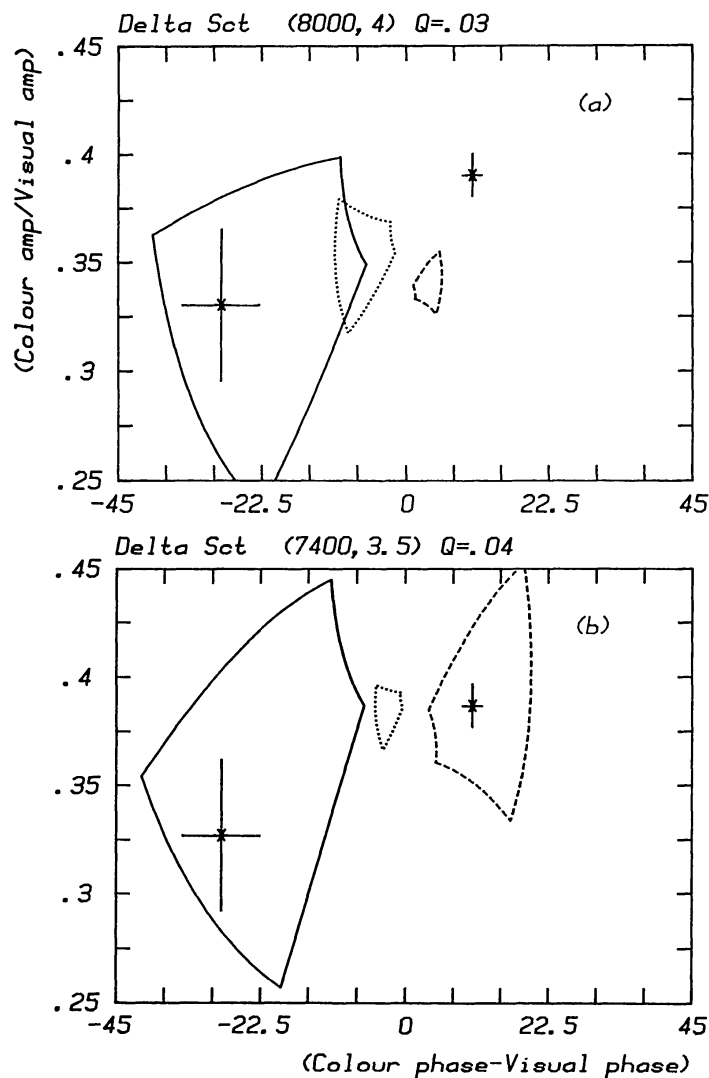


Fig. 18. Data for the star δ Scuti, from Table IV, compared with the areas of interest on the $(A_{B-V}/A_V, \phi_{B-V} - \phi_V)$ plane for δ Scuti test regimes. Dashed, dotted, and full lines correspond, respectively, to $l = 0, 1$, and 2 . Panel (a) is for the standard $(T_e, \log g, Q) = (8000, 4, 0.03)$ test regime. Panel (b) demonstrates the effect of going to parameters $(7400, 3.5, 0.04)$, which are closer to those for the star δ Scuti itself, as well as increasing $\alpha_T(B)$ by 0.3 .

5.4. RAPID Ap VARIABLES

The short periods in the rapidly oscillating Ap variables are believed to originate from high overtone p -mode pulsations of low l . Kurtz (1982b) proposed an *oblique pulsator* representation of these systems which envisages that the oscillations are symmetric about the magnetic axis.

Table V lists A_{B-V}/A_V and $(\phi_{B-V} - \phi_V)$ data for those rapidly oscillating Ap variables for which two-colour photometry of the pulsation has been published. Figure 21 displays this data on the $(A_{B-V}/A_V, \phi_{B-V} - \phi_V)$ plane, together with the corresponding δ Scuti values from Table IV for comparison. This demonstrates a clear

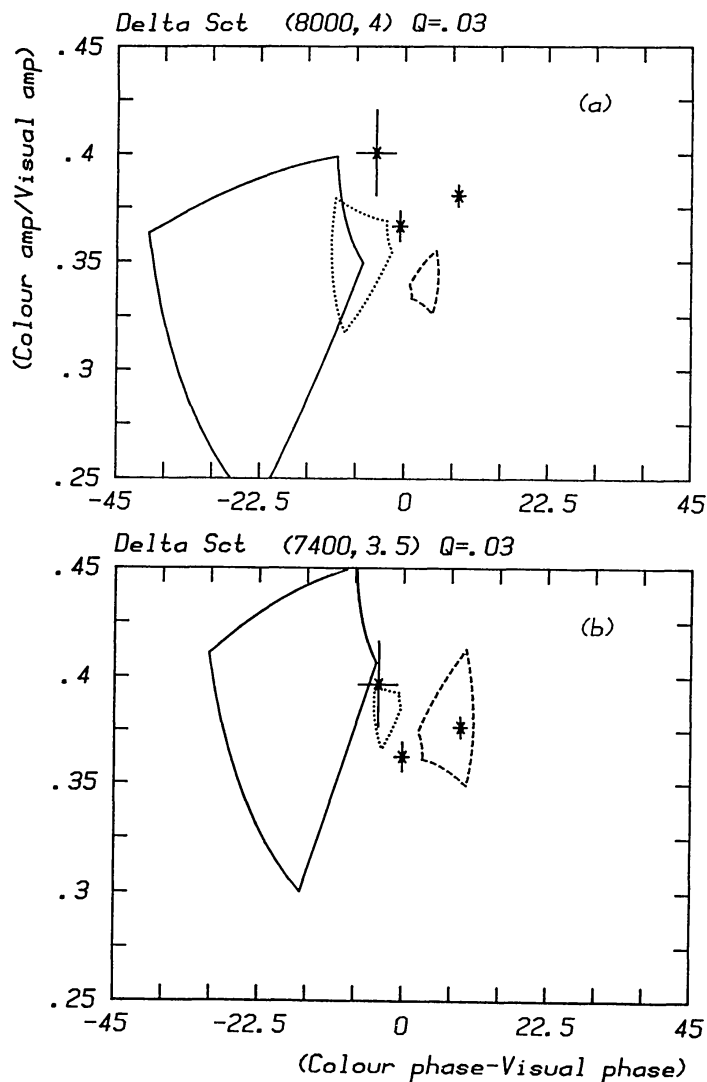


Fig. 19. Data for the star 1 Mon, from Table IV, compared with the areas of interest on the $(A_{B-V}/A_V, \phi_{B-V} - \phi_V)$ plane for δ Scuti test regimes. Dashed, dotted, and full lines correspond, respectively, to $l = 0, 1$, and 2 . Panel (a) is for the standard $(T_e, \log g, Q) = (8000, 4, 0.03)$ test regime. Panel (b) demonstrates the effect of going to parameters $(7400, 3.5, 0.03)$, which are closer to those for the star 1 Mon itself, as well as increasing $\alpha_T(B)$ by 0.3 .

difference in characteristics for the rapid Ap oscillations, even though they originate in stars with similar $(T_e, \log g)$ to the δ Scuti variables. Larger colour to light amplitude ratios are seen, and the data spread to larger negative phase differences than is the case in the δ Scuti variables. Except for BD + 28° 1494, the δ Scuti data are confined to a compact region with a mean A_{B-V}/A_V of 0.37 ± 0.08 and a mean $(\phi_{B-V} - \phi_V)$ of $+2^\circ \pm 14^\circ$. The corresponding rapidly oscillating Ap star means are 1.5 ± 0.6 and $-50^\circ \pm 35^\circ$, respectively.

In the absence of realistic stellar atmosphere parameters for the rapid Ap variables from the abundance point of view, the only possible comparison with a model is via the

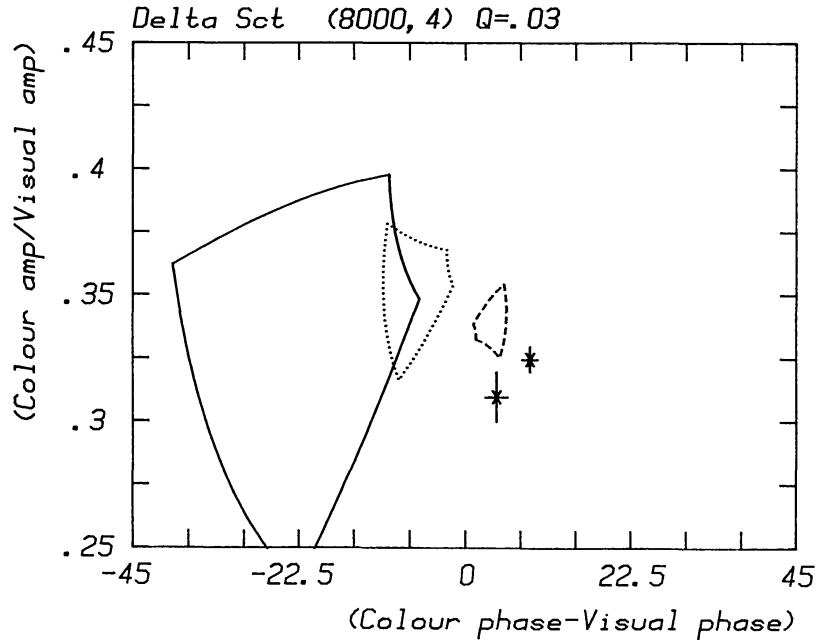


Fig. 20. Data for the star AI Vel, from Table IV, compared with the areas of interest on the $(A_{B-V}/A_V, \phi_{B-V} - \phi_V)$ plane for the standard δ Scuti test regime. Dashed, dotted, and full lines correspond, respectively, to $l = 0, 1$, and 2 .

TABLE V
Data for rapidly oscillating Ap variables

HD	P (min)	A_{B-V}/A_V	$(\phi_{B-V} - \phi_V)$ (deg)	Reference
6532	#1 6.96	1.8 ± 0.5	-97 ± 16	Kurtz and Kreidl (1985)
	#2 6.94	0.9 ± 0.3	-54 ± 20	
	#3 6.92	1.7 ± 0.8	-101 ± 27	
83368	#1 11.72	0.85 ± 0.10	-64 ± 6	Kurtz (1982b)
	#2 11.64	0.75 ± 0.15	-61 ± 8	
101065	#1 12.14	1.30 ± 0.05	-39 ± 3	Kurtz (1980a)
	#2 12.19	1.7 ± 0.6	-10 ± 28	
	#3 12.67	1.7 ± 0.6	-11 ± 24	
	#4 12.17	2.9 ± 1.5	-82 ± 40	
128898	6.82	1.30 ± 0.15	-13 ± 6	Kurtz and Balona (1984)
134214	5.65	1.05 ± 0.10	-19 ± 6	Kreidl and Kurtz (1986)

δ Scuti test regime parameters, but with the appropriately small Q value of ~ 0.0015 d. As expected from Figure 7 and the discussion in Section 4, which in this situation demonstrate predominance of terms T_1 and T_2 in (1), such a model permits only a very small region on the $(A_{B-V}/A_V, \phi_{B-V})$ plane. This region lies in the midst of the δ Scuti data very close to zero on the phase axis. It is thus spectacularly inadequate with reference to the rapid Ap star data. The difference in A_{B-V}/A_V may be explainable in

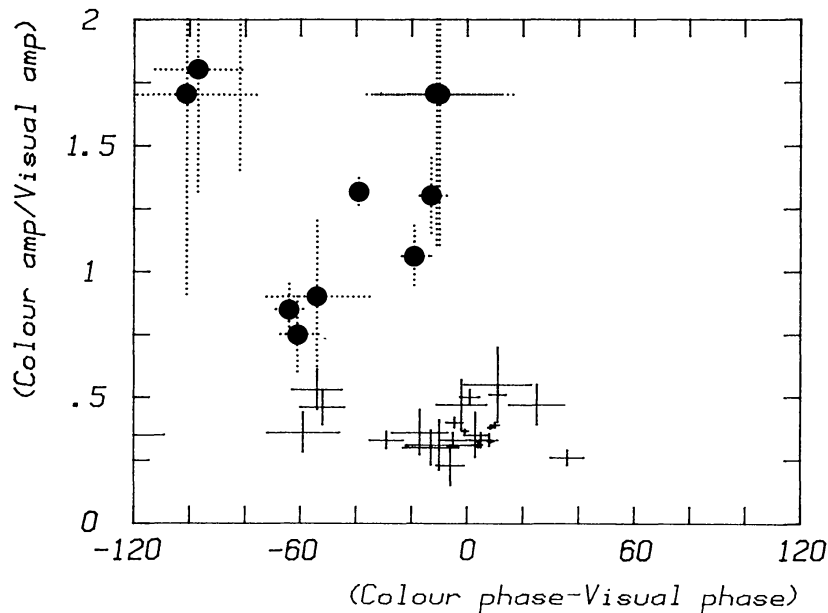


Fig. 21. A comparison of rapidly oscillating Ap star data (filled circles) and δ Scuti star data (unmarked error bars) on the $(A_{B-V}/A_V, \phi_{B-V} - \phi_V)$ plane.

terms of the abundances in the rapidly oscillating Ap systems, and the resultant effect on the $\alpha_T(B)$ and $\alpha_T(V)$ parameters in particular. In addition, the model (1) assumes uniform surface conditions, and this is certainly not the case in rapid Ap variables where both local chemical inhomogeneities and global inhomogeneities due to magnetic fields may prevail (Mathys, 1985). The restriction of the low Q model to phases near zero may suggest a more fundamental difficulty with (1). It implies that a major reduction in the relative importance of the temperature related terms in (1) is required.

5.5. CEPHEID VARIABLES

Cepheid variables are radial pulsators, but nevertheless offer a further useful comparison with (1). Such a comparison is provided in Figure 22, where, in panel (a), the areas of interest for the short-period Cepheid test regime with $(T_e, \log g, Q) = (6000, 2, 0.04)$ are shown. These areas of interest are based on the same assumptions as in the δ Scuti case. R values are assumed to lie between 1 and 0.25, and ψ_T values between 90° and 140° (Balona and Stobie, 1979b). The data plotted in Figure 22(a) are for the double-mode Cepheids studied by Balona and Stobie (1979a). Fundamental Fourier coefficients were employed for both the fundamental and first overtone periods when plotting.

Clearly, observed phases in Figure 22(a) agree with the radial mode interpretation, but the A_{B-V}/A_V values suggest, as in Section 5.3 for the δ Scuti case, that errors are present in the Kurucz-based $\alpha_T(B)$ and $\alpha_T(V)$ parameters. An increase of 0.5 in $\alpha_T(B)$ would achieve consistency of the model with the data, as demonstrated in Figure 22(d). Decreasing $\alpha_T(V)$ by ~ 0.3 would produce a similar effect. These changes are larger than any errors present in the δ Scuti regime.

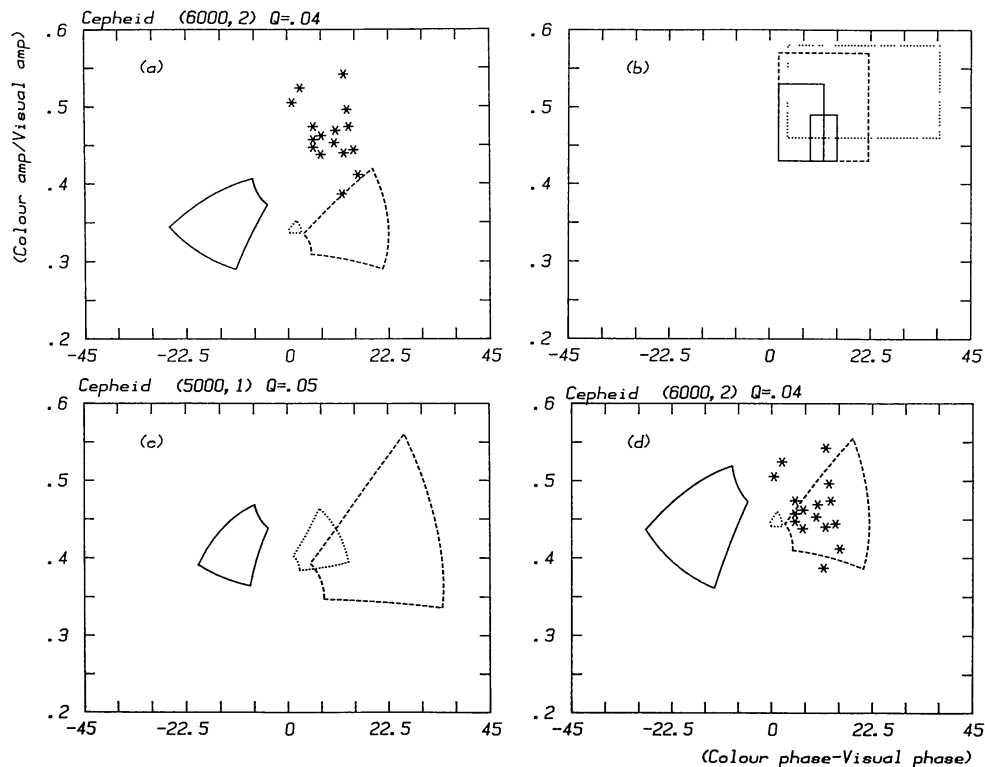


Fig. 22. Data for Cepheid variables and areas of interest on the $(A_{B-V}/A_V, \phi_{B-V} - \phi_V)$ plane. The areas of interest are based on R values of between 1 and 0.25, and ψ_T values of between 90° and 140° . Dashed, dotted, and full lines correspond, respectively, to $l = 0, 1$, and 2 . Panel (a) compares the standard short-period Cepheid test regime with data for double-mode Cepheids given by Balona and Stobie (1979a). Panel (b) gives boxes at 1σ values about the mean location of different samples of Cepheids discussed in the text. Full boxes apply to the two periods in the Balona and Stobie sample, the dashed box applies to the short-period Arellano Ferro (1984) sample, and the dotted box applies to the general Moffett and Barnes (1985) sample. Panel (c) shows the changed regions of interest that apply if a test regime more applicable to longer-period Cepheids, with $(T_e, \log g, Q) = (5000, 1, 0.05)$, is adopted. Panel (d) demonstrates that the standard short-period Cepheid test regime can be made consistent with the data if $\alpha_T(B)$ is increased by 0.5.

It appears that any differences due to intrinsic changes along the Cepheid instability region are small on the $(A_{B-V}/A_V, \phi_{B-V} - \phi_V)$ plane. This is demonstrated in Figure 22(c) where areas of interest applicable to a longer-period Cepheid regime with $(T_e, \log g, Q) = (5000, 1, 0.05)$, and Bell and Gustafsson (1978) α_T and α_g , are shown. It is also demonstrated for Cepheid observational data in Figure 22(b), where boxes at $\pm 1\sigma$ values about the mean location of different Cepheid samples are shown. The samples used are those of Balona and Stobie (1979a) for 8 double-mode Cepheids, Arellano Ferro (1984) for 9 short-period Cepheids, and Moffett and Barnes (1985) for 104 Cepheids of all periods. The mean values of $(A_{B-V}/A_V, \phi_{B-V} - \phi_V)$ are $(0.46 \pm 0.03, +12^\circ \pm 3^\circ)$ for the fundamental periods of the double-mode Cepheids, and $(0.48 \pm 0.05, +7^\circ \pm 5^\circ)$ for the first overtone periods. The corresponding means for the Arellano Ferro and Moffett and Barnes samples are $(0.50 \pm 0.07, +12^\circ \pm 10^\circ)$.

and $(0.52 \pm 0.06, +21^\circ \pm 17^\circ)$. Again fundamental Fourier coefficients of the data were used in determining these values.

5.6. ZZ CETI VARIABLES

The only ZZ Ceti system which has available a reasonable body of two-colour photometry over the pulsation is the object R548. Robinson *et al.* (1982) have determined amplitude and phase data for four periods in this object from u and y observations. Because the observational errors were substantial, they then formed a weighted mean of the data for the four periods, doing this in two different ways. The two sets of data points which result are shown on the $(A_u/A_y, \phi_u - \phi_y)$ plane in Figure 23.

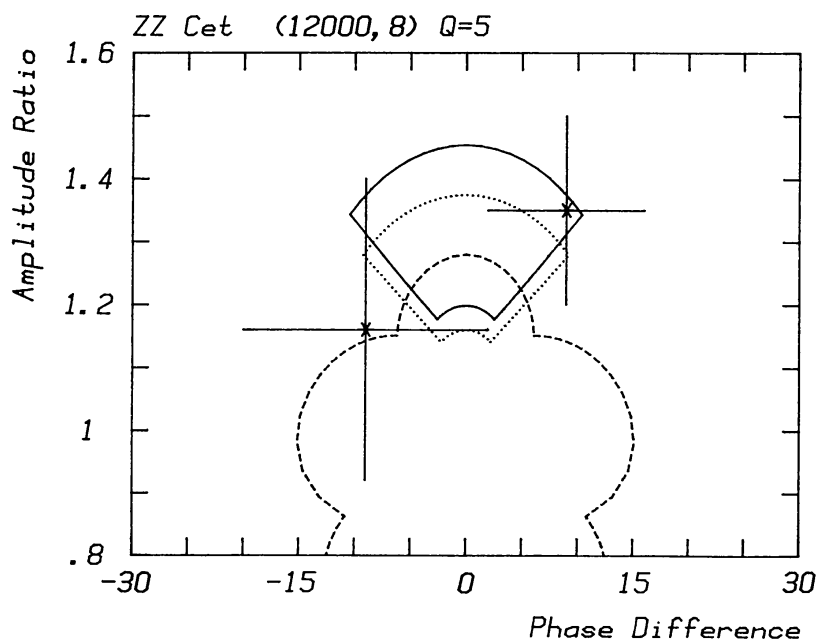


Fig. 23. Data for the star R548 compared with areas of interest on the $(A_u/A_y, \phi_u - \phi_y)$ plane for the ZZ Ceti test regime. The areas of interest are based on R values of between 1 and 0.25, and ψ_T values of between 135° and 225° . Dashed, dotted, and full lines correspond, respectively, to $l = 0, 1$, and 2 . Robinson *et al.* (1982) consider their data in two different ways, resulting in the two data points plotted on this diagram.

Figure 23 also shows, for comparison, the $l = 0, 1$, and 2 areas of interest for the ZZ Ceti test regime. These areas of interest are based on R values between 1 and 0.25 and ψ_T values between 135° and 225° . There is little evidence available to guide the selection of (R, ψ_T) ranges, other than the general expectation that white dwarf conditions may favour the adiabatic situation. Robinson *et al.* refer to a communication from Winget which implies local fractional temperature changes in appropriate models may typically vary by a factor of three from the adiabatic values. It is apparent from Figure 23 that no mode discrimination is here possible. However, bearing in mind the

uncertainties in both model and data, the assertion by Robinson *et al.*, that the two-colour data are at least consistent with low l nonradial modes, is supported. This fits with the belief that ZZ Ceti pulsations are g -modes of low l .

6. Conclusion

The relative importance of local temperature, geometry, pressure, and limb-darkening variations, in contributing to nonradial flux changes, has been documented as a function of l . In particular, attention has been drawn to the decrease in the geometry contribution which occurs if Q values are either very much larger or very much smaller than radial fundamental ones. A format for the comparison of observational light and colour data with model predictions on an amplitude ratio versus phase difference plane has been outlined. Application of this procedure to data on β Cephei, 53 Persei, δ Scuti, Cepheid and ZZ Ceti stars is generally in agreement with other information on these systems. For the rapidly oscillating Ap stars, however, the current model is inadequate.

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